From Cultivation to Care: Edge IoT, Digital Supervision and Mobile Interfaces for Medicinal-Plant Traceability

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Abstract

Medicinal-plant traceability requires links between environmental observations, material lineage, and the people responsible for cultivation and downstream medicinal-use records. We present an edge Internet of Things (IoT) architecture combining local sensing, deterministic monitoring, digital supervision, QR-based passports, and human-machine interfaces. PLANT2CLINIC provides the reference implementation: MQTT telemetry and SQLite persistence support room-state monitoring, while local traceability records and human-reviewed dossiers preserve production context. A production replica supports inspection of copied operational state. Complementary mobile demonstrators address farmer activities and healthcare research workflows; their integration boundaries and simulated inputs are stated explicitly. Local AI extends evidence retrieval and explanation without determining alarm states. Evaluation combines an audit of 55,266 stored field-monitoring evaluations, robustness tests, audit workflows, and separate evidence-assistance benchmarks. The consistency audit found no discrepancies in 136,005 recorded anomaly-band decisions or 54,982 aggregated room states, excluding 284 freshness states; it does not in-

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dependently validate anomaly detection or agronomic thresholds. A separate 100-case timing and robustness probe reported a median evaluation time of 0.24 ms, 96 controlled responses, and four input-validation exceptions. Seven audit workflows completed. The mobile interfaces illustrate complementary interactions but were not subjected to user studies or an end-to-end interoperability trial. The architecture supports a plant-to-care information pathway with human oversight and bounded AI assistance. Clinical demonstrations use fictitious data; clinical effectiveness and predictive digital-twin behaviour remain outside the validated scope.

Keywords: Internet of Things, Edge computing, Digital supervision, Human-machine interfaces, Traceability, Medicinal plants, AI assistance

1. Introduction

Background and motivation. An Internet of Things (IoT) architecture for medicinal production must connect physical observations to identifiable material and to the people responsible for interpreting and approving its records. A room dashboard alone cannot reconstruct a batch history, and a batch identifier alone cannot explain the environmental conditions associated with production. The engineering problem addressed here is how to connect sensing, edge processing, human-machine interfaces (HMIs), and traceability into a locally operable evidence pathway. PLANT2CLINIC implements this pathway from cultivation toward medicinal-use handoff, with optional AI assistance layered over deterministic monitoring and human-reviewed records.

Regulated medicinal crop production requires robust traceability, environmental monitoring, procedural consistency, and auditable decision-making. A digital system for this context must identify each plant, batch, or derived product, preserve its lineage, record operator actions, document deviations, and provide quality-oriented evidence for review. This requirement is particularly relevant when Good Agricultural and Collection Practice (GACP) principles must be translated into operational records, local workflows, and retrospective

batch reconstruction.

Digital agriculture has progressively incorporated IoT platforms, dashboards, decision-support tools, and AI-based services. However, many implementations remain fragmented. Traceability systems often focus on records and identifiers; IoT dashboards focus on telemetry; SOPs remain as static documents; and AI assistants frequently operate as external tools rather than as bounded components of a validated operational workflow. As a consequence, growers and quality managers may have access to data, but not necessarily to integrated, contextual, and auditable knowledge.

1.1. PLANT2CLINIC project context

The present work is part of the broader PLANT2CLINIC project, which aims to connect medicinal cannabis production with hospital pharmacy preparation and future clinical-trial monitoring through a continuous digital traceability pathway. PLANT2CLINIC is a collaborative effort involving three institutions with complementary expertise: (i) the *TYS (Technology and Health)* research group at the *University of Alicante* (Spain), which provides the technology platform, edge-computing architecture, and digitalisation framework; (ii) the *CONICET-CENPAT* (National Patagonian Center, National Scientific and Technical Research Council, Argentina), which contributes domain expertise on medicinal cannabis cultivation, phytochemistry, and botanical characterisation; and (iii) the *Andrés Ísola Public Hospital* (Puerto Madryn, Chubut, Argentina), which provides clinical infrastructure and expertise for the design of the double-blind randomised controlled trial module. The complete pathway includes registered genetics and mother plants, cloning, vegetative growth, flowering and harvest, extraction or concentrate preparation, quality control, storage and transport, hospital pharmacy reception, formulation or dilution, clinical dispensing, dose adjustment, and patient monitoring.

The present article does not evaluate the clinical phase and does not analyse patient data. Instead, it focuses on the agronomic and preclinical production module of PLANT2CLINIC. This module is responsible for generating the trace-

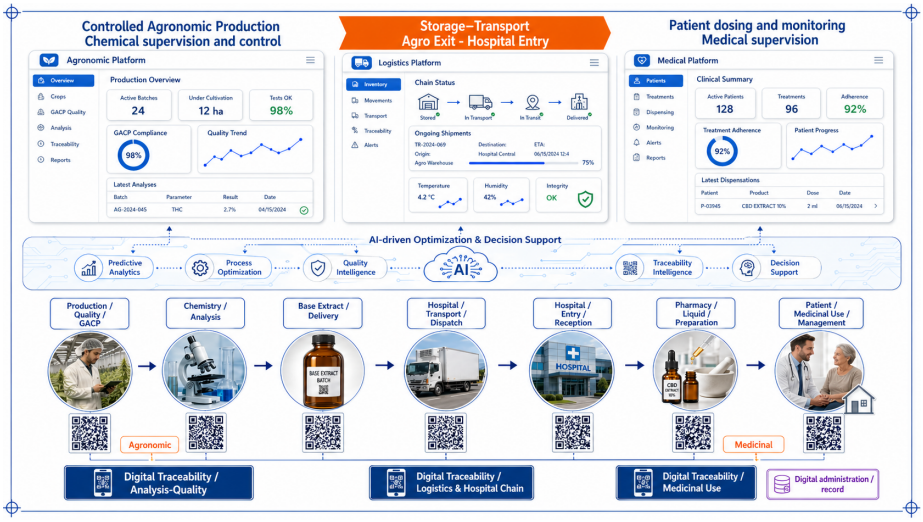


Figure 1: PLANT2CLINIC end-to-end traceability pathway and scope of the present article. The broader project links medicinal cannabis production, quality control, hospital pharmacy preparation, clinical dispensing, dose adjustment, and patient monitoring. This article focuses on the agronomic and preclinical production module, where local-first traceability, IoT monitoring, SOP evidence, and AI-assisted operational support are implemented.

ability and quality evidence required before the plant-derived material enters the hospital workflow. Therefore, the output of this prototype is not merely a crop record, but a digital dossier that can accompany the material toward hospital pharmacy reception.

The final operational goal is to deliver to hospital pharmacy a medicinal cannabis preparation supported by evidence of origin, batch lineage, environmental monitoring, SOP compliance, quality-control records, audit logs, and QR-linked traceability (see Fig. 1).

1.2. Problem statement

The practical problem addressed in this work is the disconnection between four operational layers in controlled medicinal cultivation: (i) biological and processing traceability records, (ii) environmental and fertigation IoT telemetry, (iii) Good Agricultural and Collection Practice (GACP) and Good Manufacturing Practice (GMP) standard operating procedures (SOPs), and (iv)

AI-assisted decision support. In regulated clinical-grade medicinal production, this disconnection prevents the synthesis of a reliable, integrity-verifiable chain of custody. It limits the capacity to answer operational and quality-assurance (QA) questions such as: What is the verified lineage of this medicinal batch? What room conditions were associated with its QR passport? Which specific SOP applies to a detected environmental deviation? Is an alert state derived from a deterministic policy or an AI suggestion?

A further challenge is the safe deployment of generative AI in pharmaceutical-grade workflows. Cloud-based LLM services raise data privacy concerns and introduce dependencies on external connectivity, which is unacceptable for local edge operations. Furthermore, standard RAG-based systems can hallucinate, which violates GACP/GMP validation requirements where operational truth, alarm severity, and QA releases must remain fully accountable. Therefore, the central research challenge is to design a local-first, data-localizing digital platform that integrates traceability, IoT monitoring, and local AI assistance, while enforcing a strict boundary between advisory probabilistic language generation and deterministic, human-validated operational records.

Approach. We propose PLANT2CLINIC-Agronomy, a local-first, SOP-driven cyber-physical digital platform prototype that utilizes local AI services (Llama3:8B [1] via Ollama [2]) to support clinical-level chain of custody and handoff-ready traceability dossier preparation. The platform combines QR-based batch passports, room-level IoT context, deterministic policy evaluation, local RAG-based evidence retrieval, and a dedicated cryptographic integrity layer.

The architecture is organized into three functional services: S1 as the local knowledge and evidence agent using a local LLM to explain and summarize procedures, S2 as the deterministic operational monitoring agent executing rule-based alarm policies, and S3 as the human-in-the-loop audit and QA validation layer. The S3 service acts as the human-review boundary, adding local tamper evidence by registering cryptographic integrity digests (local keccak256 hashes

of canonicalized payloads) at critical milestones, such as stage transitions, QA releases, and alarm acknowledgements. By running all services—including the S1 LLM—locally at the edge, the system keeps cultivation, telemetry and QA data on-premises and operates offline; we report this as data localization rather than as a verified privacy or confidentiality property. On notation: **S1**, **S2** and **S3** denote the three functional layers, while the suffixed **S1-1** and **S1-2** denote the two routes *within* S1—deterministic analytical retrieval and generative explanation—which differ in accuracy, latency and failure modes and must not be pooled. Tables prefixed with **S** plus a number refer to the Supplementary Material; Supplementary Table S14 lists every abbreviation used here.

1.3. Contributions

The contribution is an implemented edge IoT architecture that links room-level observations, material identity, and human review across a medicinal-plant production workflow. Its principal elements are:

- **An IoT-to-traceability evidence pathway**: MQTT ingestion, local persistence, deterministic room-state evaluation, QR-linked batch passports, and audit dossiers connect the physical cultivation environment to records that can accompany derived material toward hospital pharmacy.
- **HMIs for operational interpretation and accountability**: monitoring, passport, evidence-assistance, and QA views expose different stages of the same workflow. Sensor observations, computed alarms, generated explanations, and human decisions have distinct roles; a generated answer does not authorize an operational transition.
- **An extensible edge service architecture**: deterministic monitoring (S2) and human-reviewed audit (S3) form the operational path, while evidence retrieval and local generative explanation (S1) extend operator support. The separation allows model-dependent assistance to be assessed without treating it as the source of alarm states.

- **Component and workflow evidence**: an audit of 55,266 stored field evaluations characterizes consistency of logged anomaly decisions and room states; robustness and audit trials expose operational limitations; retrieval, model-comparison, and abstention experiments characterize the AI extension separately.

Statement of novelty. The proposed contribution lies in the integration and evaluated separation of the sensing-to-record pathway, rather than in a new sensor, communication protocol, language model, or clinical intervention. The same batch context links environmental observations, operator-facing HMIs, and human-reviewed handoff evidence. Deterministic policies retain authority over monitoring states while the optional language service explains retrieved context. The experiments characterize this implementation and its boundaries; they do not establish superiority over all existing IoT platforms. Downstream clinical modules and optional blockchain notarization remain prototype extensions of the plant-to-clinic model, not evidence of a validated clinical deployment.

Paper organization. The remainder of this paper is organized as follows. Section 2 reviews related work on digital traceability, IoT monitoring, intelligent agents, and RAG-based decision support in agriculture. Section 3 describes the materials and methods, including the local-first traceability model, IoT monitoring layer, SOP-driven knowledge base, S1–S2–S3 functional decomposition, and integrity layer; the downstream clinical-trial module extension and its three clinical agents (C-Trace, BIM, S2-C) are outside the scope of this article and are documented in the Supplementary Material as roadmap context only. Section 4 presents the local-first digital platform prototype implementation. Section 5 describes the experimental validation of the S1-1 analytical retrieval module, the S2 deterministic operational monitoring agent, and the S3 audit workflow. Section 6 reports the results. Section 7 discusses the scientific contribution, limitations, and deployment implications. Section 8 concludes the paper and outlines future work.

2. Related Work

2.1. End-to-end traceability for medicinal cannabis and plant-derived therapeutics

Traceability in regulated medicinal crop production must preserve the identity, origin, transformations, analytical results, and handling history of plant-derived material across multiple stages. For medicinal cannabis, this requirement is more demanding than conventional agricultural traceability because batch variability, phytochemical profiles, microbial safety, storage conditions, and documented handling can affect downstream pharmaceutical preparation and clinical use. Recent discussions on cannabis supply-chain traceability emphasize the need to track origin, processing, quality control, and distribution of cannabis-derived products, often proposing QR-linked or ledger-assisted mechanisms to strengthen auditability [3, 4, 5, 6, 7]. However, many traceability proposals remain focused on supply-chain transparency rather than on the operational link between cultivation rooms, sensor context, SOP compliance, and batch-level evidence.

The broader PLANT2CLINIC pathway extends this problem from agronomy to healthcare, connecting genetics, mother plants, cloning, vegetative growth, flowering, harvest, extraction, quality control, storage and transport, hospital pharmacy reception, formulation, dispensing, and patient monitoring. The present article focuses on the upstream agronomic and preclinical module, where traceability evidence is generated before the material enters the hospital workflow, and which must therefore also prepare a reliable handoff dossier for future pharmacy reception.

GACP/GMP alignment and quality evidence. Good Agricultural and Collection Practice (GACP) provides the quality basis for medicinal-plant starting materials [8]. The revised EMA/HMPC guideline emphasizes the identification of critical production steps required to ensure appropriate quality and patient safety in herbal starting materials [9]. In medicinal cannabis production, the

transition from cultivation to pharmaceutical preparation also requires attention to GMP, GLP, stability, and quality-control requirements, especially when extracts or preparations are intended for medical use [10, 11]. Recent work on cannabis stability and chemical variability further highlights the importance of linking phytochemical quality with cultivation, processing, storage, and technological drivers across the production chain [12].

These requirements motivate digital systems that connect records, procedures, environmental context, and quality evidence. A purely administrative traceability log is insufficient if it cannot explain the environmental history of a batch, the SOP context of a deviation, or the evidence supporting a release or rejection decision. For this reason, PLANT2CLINIC-Agronomy treats traceability as a structured evidence pathway rather than as an isolated list of events.

IoT monitoring and cyber-physical systems in controlled-environment agriculture. IoT platforms are widely used in controlled-environment agriculture to monitor temperature, relative humidity, light, pH, electrical conductivity, irrigation, energy use, and other variables [13, 14, 15]. Recent reviews describe smart greenhouses as cyber-physical systems in which sensors, actuators, controllers, dashboards, and AI-based services support more precise and resource-efficient production [16, 17]. Agent-based service architectures for smart greenhouses have also been proposed to move beyond dashboards by integrating telemetry analytics, decision support, RAG-grounded conversational services, and bounded operational interventions [18].

Despite these advances, most IoT deployments organize information by device, room, or time-series dashboard. In regulated production, however, quality review often requires a batch-oriented view: the question is not only what happened in a room, but which batch or QR passport was exposed to those conditions, which SOP applied, which alarm occurred, and which operator response was recorded. The contribution of the present work is therefore not the use of IoT alone, but the integration of room-level telemetry with QR-based traceability, deterministic alarm classification, and S3 audit/handoff records.

Edge computing, local-first architectures, and reduced external data exposure. Cloud platforms can simplify integration and analytics, but dependence on cloud connectivity may be problematic in operational cultivation environments, especially when data privacy, latency, cost, or resilience are important. Recent reviews of edge and cloud–edge–device computing in agriculture describe the advantages of processing data closer to the farm, including real-time response, lower bandwidth use, resilience under connectivity loss, and improved data governance [19, 20, 21, 22, 23]. Published agricultural edge systems pursue those advantages by compressing the model until it fits the device: lightweight segmentation architectures for real-time pest detection [24], knowledge distillation for livestock health monitoring [25], and low-power monitoring nodes that trade detection performance for long unattended autonomy [26]. That line of work optimises a single inference task under a hardware budget; the architecture proposed here instead partitions the workload, so that the operations requiring guaranteed response time never invoke a model at all. Deploying language models under such edge constraints in turn motivates model-compression and deployment-aware inference strategies [27]. These properties are particularly relevant for medicinal-crop production, where traceability records, environmental monitoring, QA decisions, and future clinical handoff data may be sensitive.

The local-first design adopted in this work follows this direction. IoT telemetry, traceability records, SOP retrieval, alarm classification, and local Llama3:8B-assisted evidence analysis are performed at the edge, using a Mini PC as the operational server. This differs from cloud-centric agricultural AI services by keeping sensitive operational data inside the installation while preserving the possibility of later synchronization, export, or controlled handoff.

RAG-based evidence retrieval and local AI agents in agriculture. Retrieval-Augmented Generation (RAG) [28] has become an important technique for grounding language models in domain-specific documents, mitigating the well-documented tendency of language models to hallucinate unsupported content [29]. Cost-efficient domain-specific question-answering pipelines and knowledge-

base or knowledge-graph retrieval systems illustrate how grounded LLMs can serve specialized, high-stakes domains [30, 31, 32]. In agriculture, recent surveys identify RAG and LLM-based systems as promising tools for decision support, monitoring, and knowledge access, while also emphasizing unresolved issues such as grounding quality, evaluation, domain adaptation, and safe deployment [33, 34]. Outside agriculture, industrial RAG work reaches the same diagnosis from the manufacturing side, reporting that retrieval quality governs the reliability of the generated answer and motivating structured knowledge representations to constrain it [35]. Domain benchmarks for agricultural extension question answering have begun to make this measurable [36], and vision–language models have been evaluated for crop-disease diagnosis under field rather than laboratory conditions [37]. Time-aware and domain-specific agricultural RAG frameworks further illustrate the need to align retrieved context with operational time, sensor records, and agronomic decision cycles [38]. Public-sector implementations of agricultural chat assistants have also highlighted the importance of RAG, guardrails, data governance, and policy compliance in responsible AI deployments [39].

In this article, the RAG component is not presented as an autonomous decision-maker. Instead, S1 is decomposed into query normalization, S1-1 analytical retrieval and evidence extraction, and S1-2 local Llama3:8B explanation. This design explicitly separates evidence retrieval from answer generation. The local LLM is therefore used to explain and summarize validated evidence, while operational thresholds, alarm status, traceability updates, and QA decisions remain outside generative control.

2.2. Agent-based service architectures for greenhouse AI/IoT

A particularly relevant precedent is the Agent-Based Service Architecture (ASA) for smart greenhouses proposed by Pardo-Pina et al. This work addresses a practical limitation of conventional IoT platforms: telemetry and dashboards are available, but growers often lack direct operational access to AI services that transform sensor streams into decisions, alerts, reports, and bounded in-

terventions. The ASA model organizes greenhouse intelligence into specialized services: S1, a RAG-grounded conversational agent over SOPs and local context; S2-IoT, a telemetry-analysis service for agronomic indicators and bounded actuation; and S2-E, an incident and emergency service for policy-compliant responses. Automated actions are constrained by predefined operating conditions, stage-dependent thresholds, validation gates, conflict-resolution rules, and operator-defined safety policies [18].

The ASA article is highly relevant because it provides a validated service-layer pattern for moving from raw IoT telemetry to actionable decision support. It was deployed in a 50 m^2 controlled-environment hydroponic greenhouse cultivating industrial hemp. The reported validation includes 38/40 correct answers for the S1 conversational service using GPT-5.2 (95.0%, as designated in the cited work), compared with 32/40 for GPT-4o-Mini (80.0%). For telemetry analytics, S2-IoT estimated daily light integral with MAE = 0.005 and RMSE = 0.008 mol m^{-2} d^{-1} over 13 evaluated days. The S2-E service achieved 0.81 accuracy and 0.97 stability across 36 emergency test cases. These results demonstrate that RAG-grounded agents and telemetry agents can become operational services within greenhouse IoT workflows rather than isolated chatbot prototypes [18].

PLANT2CLINIC-Agronomy builds on this service-oriented perspective but introduces a critical paradigm shift adapted to highly regulated pharmaceutical-grade production. Whereas the ASA model focuses on greenhouse control, low-code orchestration, and autonomous bounded actuation, the present work reorganizes the agent framework around a strict GACP/GMP-aligned chain of custody and plant-to-clinic evidence continuity. To achieve this, we introduce four major enhancements: (i) **Local AI Deployment**: S1 is implemented as a local-first knowledge and evidence service running a local LLM (Llama3:8B) entirely at the edge, so that cultivation, telemetry and QA data remain on-premises and the service operates offline without cloud dependencies. We describe this as data localization—an architectural property we can demonstrate—rather than as a confidentiality guarantee, which would require a formal security and privacy

evaluation that we do not claim. (ii) **Two-stage auditable RAG pipeline (S1-1/S1-2)**: Unlike the ASA model, where S1 operates as a single-step RAG-grounded conversational agent, we decompose S1 into a deterministic retrieval stage (S1-1) and a separate generative explanation stage (S1-2). S1-1 produces a structured, canonical JSON evidence block (sources, SOP versions, sensor snapshots); S1-2 uses local Llama3:8B exclusively over that block. This separation is essential for two reasons: it enforces retrieval-gated abstention (when S1-1 returns no sufficient evidence the request is resolved by a deterministic gate that emits a standardized uncertainty statement without invoking the model; Section 6.3 quantifies the boundary of this guarantee), and it generates a deterministic, canonicalizable payload that S3 can anchor by hashing—something a single-step conversational agent does not provide. (iii) **S3 Integrity Layer**: We design and incorporate a dedicated S3 audit and QA governance agent. Unlike prior models that rely on simple logging, the S3 agent registers cryptographic integrity digests (keccak256 hashes of canonicalized batch payloads, where object keys are sorted before serialization so that semantically identical records hash identically) at critical milestone points (e.g., stage transitions, alarm acknowledgements, and quality-control releases), so that any later modification of a sealed record becomes detectable by recomputation. This is a tamper-*evident* mechanism, not a tamper-proof one: the hashes are stored locally and detect alteration rather than prevent it. A tampering benchmark executed against the production integrity module detected 29 of 29 alterations spanning field mutation, deletion, reordering, replay and concurrent anchoring, while correctly not flagging 5 semantically equivalent transformations; the single case it does not detect — an adversary who rewrites the co-located anchor store itself — is reported as a declared boundary of the mechanism rather than omitted (Supplementary Table S11 and Fig. S2, Integrity-Layer Tampering Benchmark section). The stronger immutability guarantees of an append-only distributed ledger are outside the scope of this deployment (Section 7). (iv) **Handoff-ready traceability dossier**: S1 evidence, S2 deterministic alarms, and operator responses are consolidated into a QR-linked, cryptographically an-

chored digital dossier at the edge, so that a validated compliance history—not raw telemetry—accompanies each batch to the next stage of the supply chain (downstream, clinical pharmacy reception).

Comparability of architectural evidence. The ASA greenhouse study [18] provides a relevant example of coordinated AI/IoT services. Its model-based incident tests and our audit of logged deterministic decisions answer different questions, so their percentages are not treated as a head-to-head accuracy comparison. The relevant design distinction is the location of generative inference: in PLANT2CLINIC, explicit room-state policies and structured record lookup remain outside the language-model route. The evaluation therefore separates record consistency, robustness, and explanation quality rather than reducing the system to a single accuracy score.

Agentic decision support and human-in-the-loop control. Recent agentic AI and IoT frameworks describe coordinated perception–decision–action loops for precision agriculture, integrating sensors, analytics, recommendations, and actuation [40]. Such approaches are promising, but in regulated medicinal production they must be bounded by deterministic policies, validation gates, audit logs, and human accountability. The risk is not only technical failure but also the conversion of fluent AI-generated text into an apparent operational fact.

The S1–S2–S3 decomposition used in PLANT2CLINIC-Agronomy addresses this concern: S1 retrieves and structures evidence, S2 classifies operational states using deterministic policies, and S3 provides human-in-the-loop audit and handoff preparation. Unlike systems that treat the chatbot as the operational authority, here AI assists, deterministic rules classify, and humans validate.

Positioning relative to prior systems. Table 1 positions the proposed platform relative to representative directions in digital agriculture. The comparison is qualitative because experimental setups, crops, regulatory settings, and evaluation protocols differ across studies. The distinctive feature of PLANT2CLINIC-Agronomy is the integration of QR-based traceability, room-level IoT context,

S1 local evidence retrieval, S2 deterministic operational monitoring, S3 human-in-the-loop audit and handoff preparation, and local-first operation that reduces external data exposure, within a single agronomic module designed for a broader plant-to-clinic pathway.

2.3. State-of-the-art synthesis and research gap

The reviewed literature shows a clear progression from isolated sensing and dashboard systems toward integrated cyber-physical platforms combining IoT, edge processing, AI services, and user-facing decision support. Four trends are especially relevant. First, controlled-environment agriculture increasingly relies on heterogeneous IoT infrastructures that monitor climate, fertigation, lighting, energy, and crop status. Second, RAG and LLM-based assistants are emerging as natural interfaces for operational knowledge, SOPs, manuals, and historical data. Third, agent-based service architectures show how advisory services, telemetry analytics, and emergency services can be orchestrated under explicit policies. Fourth, local-first and edge-oriented designs are becoming important for resilience, privacy, and operational continuity.

However, important gaps remain for regulated medicinal cannabis production. Most smart-greenhouse systems are organized around rooms, sensors, dashboards, or control loops, while medicinal production requires batch-oriented evidence continuity. Most RAG or chatbot systems focus on answering questions, but do not necessarily generate a structured evidence block linked to QR passports, SOP versions, IoT snapshots, and audit records. Most agentic IoT systems emphasize monitoring or actuation, while regulated workflows require a separate layer for QA validation, corrective records, and accountability. Finally, few systems explicitly prepare an agronomic digital dossier that can accompany plant-derived material toward downstream pharmacy or clinical workflows.

PLANT2CLINIC-Agronomy addresses this gap by positioning the agronomic installation as the first validated module of a broader plant-to-clinic traceability pathway. Its contribution is not only the deployment of IoT, RAG, or local AI, but the integration of these components into an auditable evidence pipeline.

S1 converts local SOPs, documents, traceability records, and IoT context into structured evidence; S2 classifies operational states through deterministic policies; and S3 consolidates human-reviewed decisions, QA records, QR passports, and handoff evidence. This combination differentiates the proposed platform from conventional traceability tools, IoT dashboards, and generic AI chatbots.

3. Materials and Methods

This section describes the architecture and design principles of the proposed prototype. Following a layered structure similar to smart greenhouse service architectures, the system is organized into traceability, IoT monitoring, policy evaluation, knowledge retrieval, agent services, user interfaces, and optional integrity evidence.

Prototype overview. A local-first digital platform prototype was implemented to validate the proposed architecture in an operationally realistic environment. The prototype integrates traceability management, QR-based batch passports, IoT room monitoring, alarm visualization, SOP-based assistance, and local audit mechanisms.

The platform follows a local-first design. Traceability records are maintained in a local mirror, while IoT telemetry is persisted in a local SQLite database [41]. Communication between sensor gateways and the backend uses the MQTT protocol [42]. The user interface exposes the system through dedicated modules, including dashboard, QR passport viewer, IoT data view, assistant, actuator interface, audit view, and backup/restore utilities. The overall architecture of the prototype is shown in Fig. 2. A representative physical deployment of the prototype in an indoor cultivation scenario is presented later in Fig. 10.

3.1. IoT architecture and plant-to-clinic information flow

The architecture separates five responsibilities: physical sensing and gateway communication; edge ingestion and persistence; deterministic interpretation of

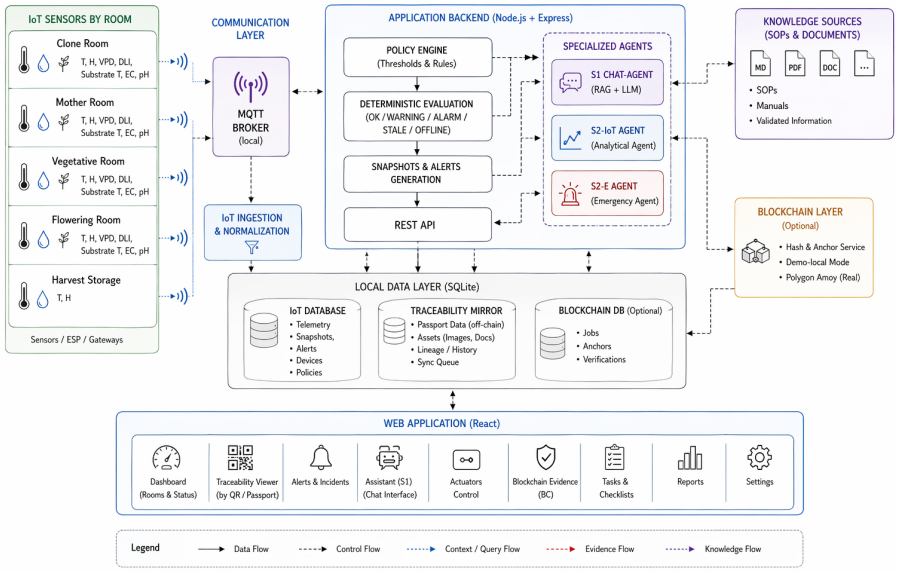


Figure 2: Overall architecture of the proposed cyber-physical traceability platform prototype. The platform integrates local-first traceability, IoT ingestion, deterministic policy evaluation, SOP-driven knowledge, local AI-assisted evidence analysis, policy-based operational monitoring, and an optional integrity layer within a unified platform interface.

room conditions; human interaction and traceability review; and optional AI-assisted access to evidence. MQTT transports measurements to the local backend, while web interfaces expose persisted observations, computed states, and batch context. The biological material and its digital passport follow linked production stages; the telemetry provides room-level environmental context rather than a direct measurement of medicinal efficacy or product composition.

The information flow is sensor/gateway → MQTT ingestion → timestamped local readings → room-state policy → HMI inspection → QR-linked audit and handoff dossier. This is an evidence and supervision pathway, not a demonstrated closed-loop actuation controller. Operator actions and QA decisions are represented explicitly in the audit workflow. S1 assistance can explain retrieved evidence alongside that pathway, but does not replace its policy engine or human approval boundary.

Three design boundaries govern interpretation. First, a current room snapshot supplies context and must not be interpreted as proof of a batch's complete lifetime exposure. Second, a reproduced threshold decision establishes implementation fidelity, not the scientific validity of the threshold or sensor calibration. Third, downstream pharmacy and clinical views demonstrate continuity of identifiers and records on simulated clinical data; they do not establish clinical interoperability or patient benefit.

Persistence and deployment domains. The central application persists traceability snapshots, assets, queue state, and history in a local filesystem mirror; its IoT histories, snapshots, and alerts are stored separately in SQLite. The production replica copies these artifacts and does not create an independent source of sensor truth. The farmer demo stores its working state in browser local storage, with an optional Google integration. Health2Clinic persists its records in its own server-side SQLite database. Thus "local-first" describes the central operational architecture and the farmer's device mode; it is not a claim that the hosted healthcare demo works offline or that all applications share a database. These boundaries matter because a copied record, an observation timestamp, and a human-entered event have different provenance and freshness semantics.

Local-first traceability layer. The traceability layer is organized as a sequence of linked production nodes. Each node represents a traceable entity, such as a genetic record, mother plant, clone batch, vegetative batch, flowering batch, harvest batch, extraction record, or derived product. Each node stores its own identifier, type, data fields, immediate parent, synchronization status, and audit metadata.

The QR passport is the operational view of a node. It reconstructs the current record, its immediate origin, and the full lineage path (see Fig. 3). This allows an operator or quality reviewer to scan any QR code and reconstruct the retrospective history of the batch or product. Importantly, the lineage is

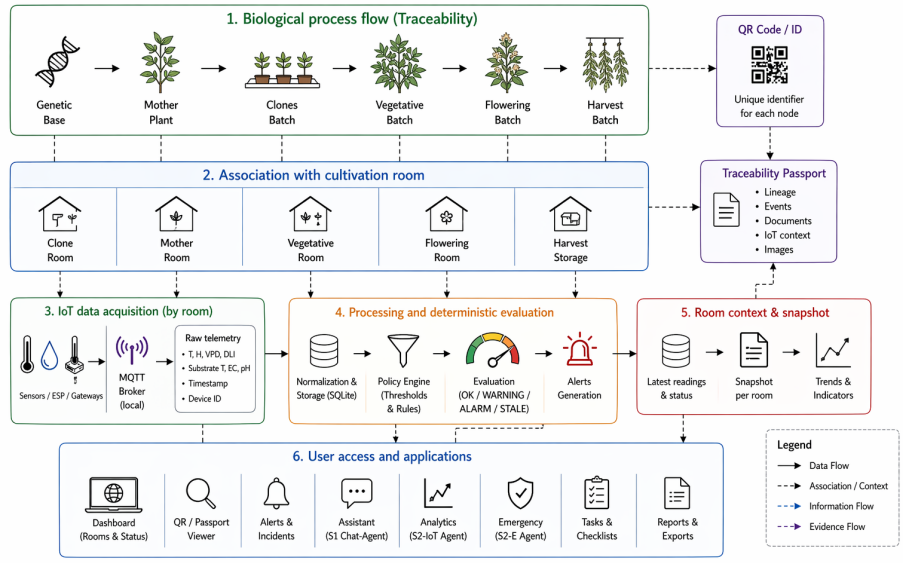


Figure 3: Traceability and IoT contextualization workflow. Biological and operational lineage is represented through QR-based passports, while room-level IoT snapshots provide environmental context for each traceable node.

not treated only as a string pattern; the system prefers explicit parent-child relationships between records.

IoT monitoring layer. The IoT layer captures telemetry from sensors and gateways using a local MQTT broker. The backend normalizes incoming payloads, validates timestamps and values, persists readings in SQLite, and generates room-level snapshots. The monitored areas include cloning, mother plants, vegetative growth, flowering, and post-harvest storage. Cultivation rooms include variables such as temperature, relative humidity, VPD, DLI, substrate temperature, EC, and pH, whereas post-harvest storage focuses on temperature and humidity. The deterministic data pipeline is illustrated in Fig. 5.

Deterministic policy engine. The policy engine evaluates readings against room-specific rules, generating operational states (OK, WARNING, ALARM, STALE, OFFLINE) from explicit thresholds, freshness rules, and anomaly policies. This

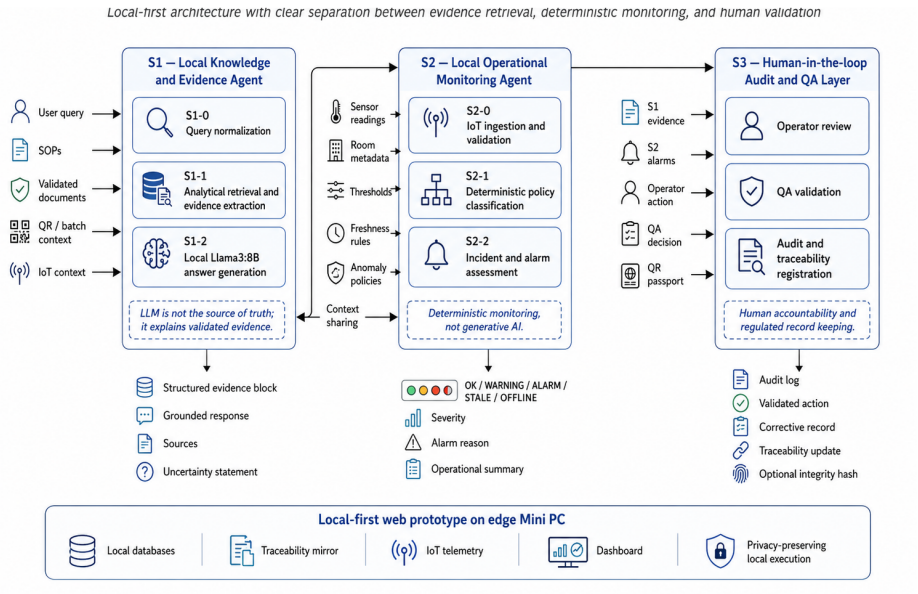


Figure 4: Functional decomposition of local decision-support services. S1 retrieves and structures evidence, S2 validates telemetry and classifies operational states, and S3 provides human-in-the-loop audit, QA validation, and traceability registration within the local-first digital platform prototype.

deterministic classification is central to the safety rationale of the prototype: alarm status and severity are never generated by the LLM.

SOP-driven knowledge layer. The system uses SOPs and validated documents as controlled sources of operational knowledge. These include traceability procedures, IoT monitoring procedures, cloning workflows, extraction workflows, quality-control procedures, and assistant-use procedures. SOPs define what the assistant may retrieve, what the operator should verify, and which procedural boundaries apply to each process.

RAG implementation parameters. Table 2 summarises the key technical parameters of the S1 RAG pipeline as deployed in the prototype. Documents are indexed at startup from local directories (SOPs, validated Markdown files, PDFs,

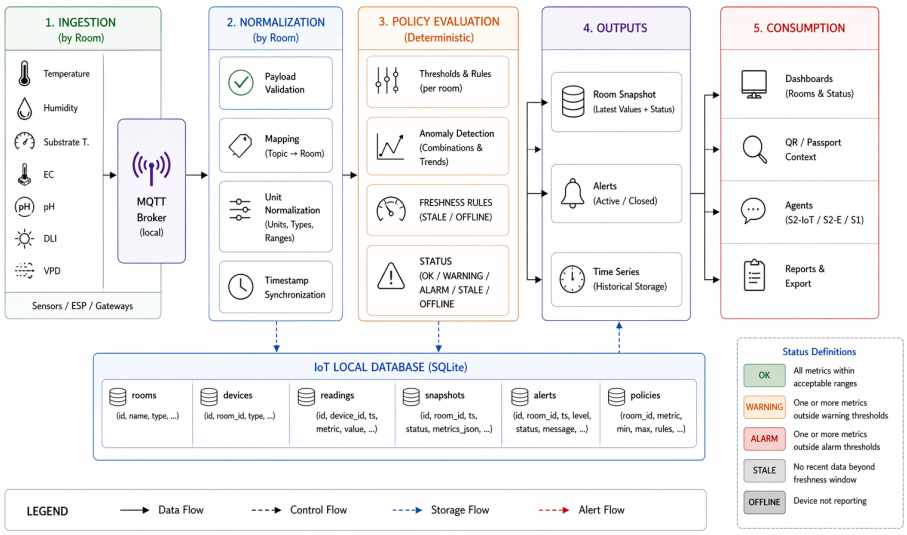


Figure 5: Deterministic IoT data pipeline. Sensor readings are ingested through MQTT, normalized, stored locally, evaluated against room-specific policies, and exposed through snapshots, alerts, dashboards, QR passports, and agent services.

and JSON records); the knowledge base comprised 11 source documents totalling approximately 47 000 words, extracted with *pdf-parse* for PDFs and read directly otherwise. Each document is split into fixed-size character chunks with a sliding overlap window, and one embedding per chunk is computed locally with *nomic-embed-text* via Ollama, without any external API call. At query time, only chunks whose cosine similarity to the query exceeds the threshold are returned, up to the retrieval limit, so the structured evidence block passed to S1-2 contains only the highest-scoring, threshold-filtered context that constrains what the local Llama3:8B model can draw on.

3.2. Functional agent decomposition

The proposed prototype separates decision-support responsibilities into three functional layers rather than treating all intelligent behavior as a single chatbot or generic AI component. This decomposition is central to the safety and

auditability rationale of the system.

S1 – Local Knowledge and Evidence Agent is responsible for retrieving, filtering, and structuring validated knowledge before local language generation. It operates over SOPs, validated technical documents, genetic information, QR/batch context, and IoT snapshots. Internally, S1 is decomposed into: S1-0 query normalization, S1-1 analytical retrieval and evidence extraction, and S1-2 local Llama3:8B answer generation. S1-1 itself resolves a query through one of two non-generative submodes, and the distinction matters for interpreting the results reported in Section 6. **S1-1a, structured field lookup**, answers queries that map onto typed records of the traceability store (genetic parameters, batch attributes, room state) by deterministic pattern matching over normalized query text, returning the stored value directly; no embedding model and no language model participate, which is why this submode resolves in a median of 31 ms and is bit-identical across language back-ends. **S1-1b, vector retrieval**, answers document-grounded queries by applying a *neural embedding model* (`nomic-embed-text`, served locally via Ollama) to compute cosine similarity between the query and indexed SOP chunks, ranking existing passages without generating text. Both submodes emit the same structured, canonical JSON evidence block. S1-2 then applies the *generative* local Llama3:8B model exclusively over that block. The invariant is that only S1-2 generates natural language; S1-1 either looks up or ranks content that already exists. Throughout Sections 5.1–6, the 100-query evaluation bank exercises S1-1a on its 36 structured-field queries; the vector retriever S1-1b is evaluated separately as an isolated component with ranking metrics (Section 6.2 and Supplementary Table S4).

S2 – Local Operational Monitoring Agent is responsible for telemetry validation and deterministic operational-state classification. It operates over sensor readings, room metadata, thresholds, freshness rules, and anomaly policies. Internally, S2 is decomposed into: S2-0 IoT ingestion and validation, S2-1 deterministic policy classification, and S2-2 incident and alarm assessment. Although S2 does not use generative AI, it is treated as an agentic compo-

nent because it implements a perception–evaluation–classification loop over IoT telemetry. Its behaviour is policy-based and deterministic: it validates sensor inputs, applies room-specific rules, assigns operational states, and emits alerts or summaries. This design preserves interpretability and prevents LLM-generated outputs from defining alarm severity or operational status.

S3 – Human-in-the-loop Audit, QA, and Hospital Handoff Layer represents the validation and accountability layer. It receives S1 evidence, S2 alarms, QR passports, operator actions, and QA decisions. Its outputs include audit logs, validated actions, corrective records, a structured S3 audit record, and optional integrity evidence. In the current prototype, the handoff output is implemented as a handoff-ready exportable dossier rather than as a live hospital information-system integration. S3 is not a generative AI service; it is the layer where human accountability, QA review, and auditability are enforced. The functional decomposition of these services is shown in Fig. 4 and summarized in Table 3.

A critical architectural rule is that local AI assists interpretation but does not execute or validate operational truth. The Llama3:8B model is restricted to the S1-2 explanation layer and operates strictly over the structured evidence prepared by S1-1 (see Fig. 6). It does not define IoT thresholds, generate alarm severity, write the traceability mirror, consolidate queues, or control actuators.

This explicit separation between S1-1 (analytical retrieval) and S1-2 (generative explanation) is essential for three reasons:

- **Hallucination Containment**: Under GACP/GMP validation guidelines, process guidance must be fully grounded. S1-1 applies a deterministic relevance threshold to the retrieved SOP fragments; if no matching criteria are met, the workflow declares uncertainty rather than delegating the question to the local LLM in S1-2. The guarantee is bounded and we state its scope explicitly: it covers the case in which no evidence is retrieved, which the gate handles deterministically, but it does not cover the case in which evidence is retrieved and is misleading. Section 6.3 reports the abstention battery that

quantifies both regimes.

- **Cryptographic Audit Trail**: The S3 agent anchors operational decisions by hashing the exact evidence used. We say *hashing* rather than *signing* throughout: these are unkeyed digests, so they bind a record to its content but carry no assertion of authorship, and their threat model is detection of later alteration by a party who does not also control the local digest store—not prevention of alteration, and not non-repudiation. Because S1-1 outputs a structured, canonical JSON payload (detailing exact document versions, paragraph hashes, and sensor snapshots), S3 can compute and register a reliable keccak256 digest of that canonicalized payload. A single-step chatbot model (only S1) would produce unstructured text, which cannot be systematically verified or audited.

- **Edge Computing Efficiency**: Running local LLMs (Llama3:8B) on a local Mini PC is constrained by memory and context length. S1-1 performs fast, lightweight local database lookups (S1-1a) and vector queries (S1-1b) first, filtering out noise and passing only the highly relevant context to S1-2. This minimizes the LLM prompt size; the deterministic S1-1a lookup route resolves in a median of 31 ms, while the bounded generative S1-2 route completes in a median of 18.8 s per query on the edge host (Section 6.5).

This separation preserves auditability and reduces the risk of probabilistic outputs affecting regulated operations.

Agent autonomy and safety boundaries. In this work, the term agent does not imply that all services are based on generative AI. S1 includes a local LLM-based RAG component; S2 is a deterministic policy-based operational agent; and S3 is a procedural human-in-the-loop audit layer, not an AI agent in the predictive or generative sense. The autonomy levels and safety boundaries of these layers are summarized in Supplementary Table S1. The agentic nature of the architecture derives from the decomposition of sensing, evidence handling, classification, and validation responsibilities into coordinated services.

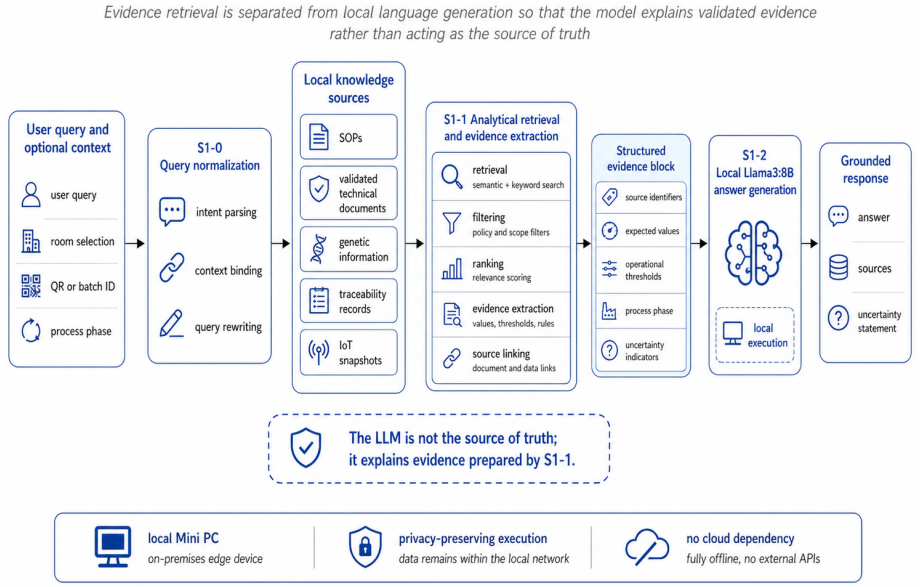


Figure 6: S1 local analytical RAG pipeline. S1-0 normalizes the query and context. S1-1 resolves the query without generating text, either by deterministic structured field lookup (submode S1-1a, used by the 36 structured-field queries of the evaluation bank) or by vector retrieval over indexed SOP chunks with a *non-generative* neural embedding model (submode S1-1b, `nomic-embed-text`, cosine similarity); both return the same structured, canonical JSON evidence block. S1-2 then uses the *generative* local Llama3:8B model exclusively over that evidence block to produce a grounded natural-language response for the operator. The LLM explains validated evidence rather than acting as the source of truth; when S1-1 retrieves no validated documentation, a deterministic gate serves the request and S1-2 is not invoked.

No LLM-generated output can directly modify traceability records, alarm thresholds, QA status, or actuator states. S1 may explain evidence, S2 may classify operational status through deterministic policies, and S3 converts reviewed evidence and actions into accountable records.

Integrity and evidence layer. The prototype includes an optional integrity and evidence layer designed to reinforce traceability without altering the operational core of the platform. This layer is not the foundation of the system and must not be interpreted as a blockchain-based traceability architecture. The core trace-

ability, IoT monitoring, and AI-assisted interpretation remain fully functional without external blockchain availability.

3.3. Integrity layer and digital notarization

The integrity layer provides a mechanism to fix operational records at specific milestones without requiring a continuous blockchain connection or storing sensitive data on-chain. As shown in Fig. 7, the system can select a QR or batch artifact, reconstruct a canonical payload, compute a keccak256 hash of that canonicalized payload and store this digital fingerprint locally. When anchoring on an Ethereum-compatible network such as the Polygon Amoy *testnet* (used here strictly as a non-production proof-of-concept, with no economic value and no guarantee of persistence), the same keccak256 digest required by the EVM is submitted, so local and external anchors are byte-identical and directly comparable; no separate hash function is used for the local path.

In the proposed architecture, this layer is used for automatic notarization at critical milestone points:

- **Stage Transitions**: When a batch moves from vegetative to flowering, or from harvest to extraction.
- **QA Release/Rejection**: Fixing the state of a batch after a quality-control decision.
- **Critical Alarm Acknowledgment**: Anchoring the evidence block (S2 alarm + S1 SOP context + operator response) to prevent retrospective modification of incident records.

This approach supports a "Trust-but-Verify" audit model. An auditor can recalculate the hash of any current record and compare it against the local anchor log. If the hashes match, the record is verified as intact. This local notarization provides a safety boundary against data tampering within the edge environment, while the optional external anchoring (e.g., on Polygon Amoy) serves as a proof-of-concept for multi-party verification.

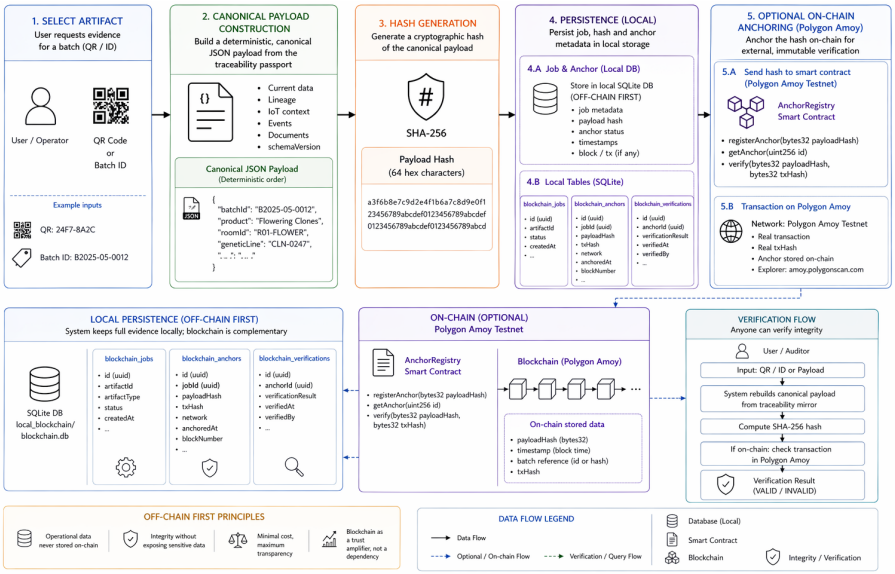


Figure 7: Integrity and evidence layer for traceability. The system selects a QR or batch artifact, constructs a canonical payload, generates a keccak256 hash, persists the job and anchor locally, and optionally anchors the hash on an external verification network, such as Polygon Amoy, as an implementation example. This layer is complementary and does not replace the local-first traceability system.

4. Platform Prototype Implementation

4.1. Connected supervision and two mobile demonstrators

The reference ecosystem comprises the central PLANT2CLINIC production application, its replicated supervision node, and two complementary mobile demonstrators: cannabisIA for farmers and Health2Clinic for authorized healthcare personnel. These components address different responsibilities along a proposed cultivation-to-care information pathway (Figure 8). Their coexistence does not establish a fully integrated clinical deployment. In particular, neither a shared identifier namespace nor automatic exchange between the mobile demos and the production system was validated in the reported experiments.

Digital-twin supervision and replica boundary. The project uses the term *digital twin* for the node connected to production traceability, IoT records, and process views. The documented replication procedure copies production state to a secondary node and disables its direct MQTT ingestion. This supports inspection of a copied operational representation without treating the replica as a second sensor authority. Replication is scheduled or manually triggered; no measured synchronization lag, lossless failover, predictive crop model, or bidirectional physical synchronization is reported here. Literature distinguishes digital models, shadows, and twins by their integration level [43]; accordingly, this article uses *digital supervision* for the demonstrated capability and separates it from a predictive or fully bidirectional digital twin. The deployment replica and the room-telemetry view are related but distinct architectural elements.

Farmer mobile HMI. The public cannabisIA demonstrator supports navigation from a lot to its phase, activities, history, and QR passport. The inspected interface offers local device persistence and optional user-authorized Google Sheets storage; the latter was not connected or evaluated in this study. Its IoT view explicitly labels its readings as simulated, and its AI view offers three prepared examples. These views illustrate how sensing and evidence assistance can be exposed to a farmer, while the real telemetry and local model benchmarks belong to the central PLANT2CLINIC stack. The five public video demonstrations cover field registration, mobile traceability, optional IoT adaptation, optional GACP assistance, and local regulatory context (Supplementary Demonstration Guide).

Healthcare mobile HMI. Health2Clinic provides a separate browser–server application for protocol-defined research workflows. The documentation describes a Node.js HTTP server and SQLite storage for accounts, sessions, protocols, coded participants, visits, assignments, product units, and activity records. The healthcare view exposes participant interviews and follow-up; the industry role manages product-related operations. The live inspection used the supplied healthcare role and confirmed the participant overview and blank first-interview

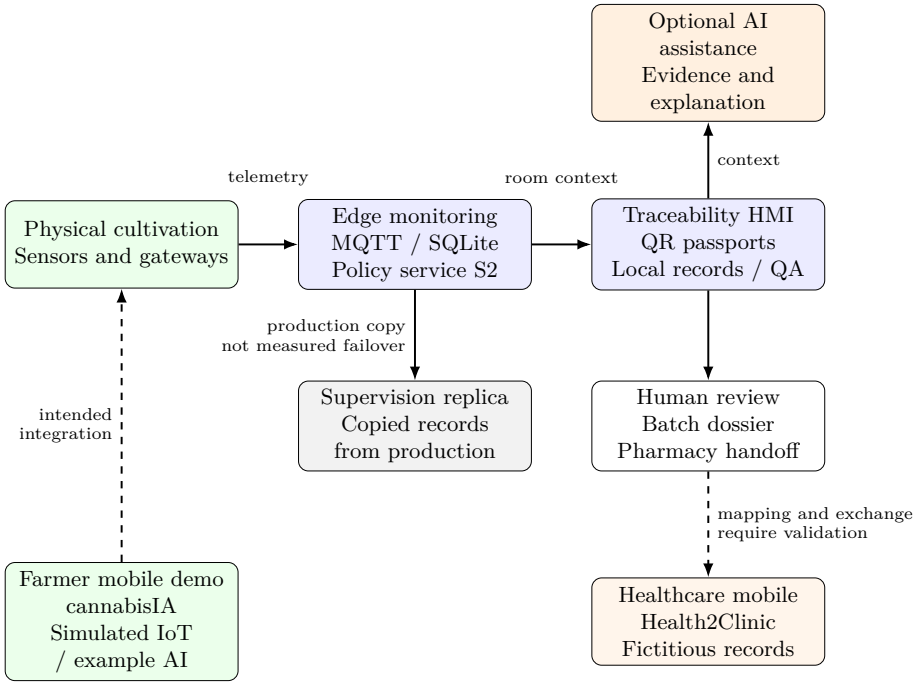
form, including an external-questionnaire response sheet. No participant was created, no product was dispensed, and no clinical decision was made. The interface states that it is a test environment with fictitious data and does not recommend doses. Documentation of other roles is treated as documented functionality, not as a live test of those roles.

Health2Clinic must be distinguished from the earlier PLANT2CLINIC RCT prototype described in the Supplementary Material. Features of that prototype—including its clinical agents and dual-authorization unblinding—are not automatically attributed to Health2Clinic. The 0.2.1 technical dossier and the current 0.3.0 training manual document different releases; claims in this paper identify that provenance and do not assume identical implementation across releases.

Figure 9 illustrates the two interfaces. The common architectural idea is that people interact with observations and traceable entities through bounded views. Implementing the proposed cross-application handoff would require a verified mapping between production batch, preparation/product unit, and clinical record, together with event times, consent and access controls, and explicit reconciliation rules. That integration remains an engineering and validation task rather than an inferred consequence of the demonstrations.

Traceability passport and QR workflow. Each traceability node can be accessed through a QR code or identifier. The passport view displays the node data, origin, lineage, documents, images, audit metadata, and associated IoT room context. The passport also acts as the input artifact for optional integrity evidence generation.

IoT dashboard and room-level monitoring. The IoT interface displays room status, latest telemetry, active alerts, and freshness information. The system differentiates between cultivation rooms and post-harvest storage because their relevant variables differ. Operators can inspect current status and quality personnel can review environmental context associated with a batch.



Solid: documented central path and replica copy.
Dashed: proposed cross-application integration, not an evaluated connector.

Figure 8: Architecture and integration boundaries of the cultivation-to-care model. Solid arrows represent the documented central data path and production-to-replica copy. Dashed arrows indicate intended exchange with separately deployed mobile demonstrators or downstream workflows, not a tested interoperability link. The AI service is advisory. This explanatory diagram was prepared with AI-assisted LaTeX drafting; it is not an experimental image.

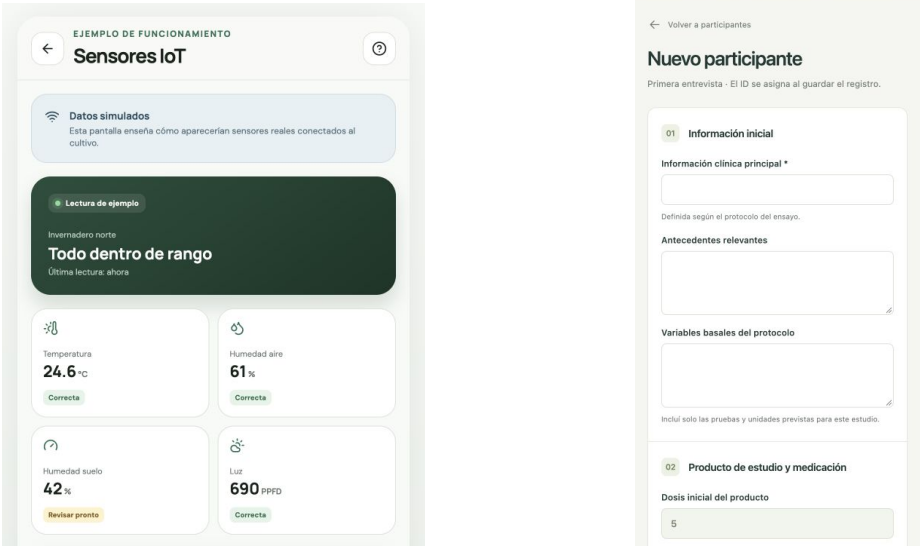


Figure 9: Complementary HMI demonstrations inspected on 16 October 2026. Left: the farmer IoT interface explicitly labels its values as simulated. Right: the mobile healthcare first-interview form in the fictitious-data environment, before any entry was saved. The screenshots illustrate role-specific interaction and were not used as measurements of usability, sensor performance, or clinical outcomes. The original interface language is Spanish; the farmer screenshot is cropped only to remove surrounding blank margins.

4.2. Human-machine interfaces for IoT supervision and traceability

The frontend interface of the PLANT2CLINIC-Agronomy prototype is implemented as a lightweight, local-first web application that serves as the visual representation of the agricultural twin node (see Supplementary Fig. S1; the web application design—module structure, view–agent mapping, and how the interface materialises the advisory/deterministic boundary—is described in the Supplementary Material, Prototype Web Interface section). The current central web frontend uses React and Vite, with the Node.js/Express backend serving the application and its local API. The mobile demonstrators are separately deployed interfaces, not evidence that all components execute in one browser process or share a database. The interface can be served by the local edge installation. The service timings reported in Section 6.5 should not be interpreted as a browser-rendering or HMI usability benchmark.

Table 1: Positioning of PLANT2CLINIC against representative published systems, replacing generic system categories with concrete comparators: ASA, an agent-based service architecture for smart greenhouses with RAG-grounded LLM agents [18]; TARAG, a time-aware agricultural RAG framework [38]; AgPal, a public-sector agricultural chat assistant [39]; AgIoT, an agentic AI/IoT precision-agriculture framework [40]; and Ledger, ledger-based cannabis supply-chain traceability [3]. Marks reflect only what each cited publication reports: ✓ reported, – not provided by design or explicitly out of scope, n/r not reported (not evidence of absence), and n/a not applicable to that system class. The downstream clinical modules of PLANT2CLINIC are roadmap context and are excluded from this comparison.

Dimension	**ASA**	**TARAG**	**AgPal**	**AgIoT**	**Ledger**	**This work**
Traceability and operational record						
Batch-level lineage with QR passport	–	–	–	–	✓	✓
Room-level IoT telemetry context	✓	✓	–	✓	–	✓
Alarm/incident classification is deterministic	–[a]	–	–	n/r	–	✓
Language-model discipline						
RAG-grounded domain assistant	✓	✓	✓	n/r	–	✓
Generative path barred from writing the record	–	n/r	n/r	n/r	n/a	✓
Model-independence of the record path quantified	–[b]	n/r	n/r	n/r	n/a	✓
Controlled retrieval ablation as causal test	–	n/r	n/r	n/r	n/a	✓
Deployment and compliance						
Fully local (on-premise) inference	–	n/r	n/r	n/r	n/a	✓
Per-milestone cryptographic integrity sealing	–	–	–	–	✓[c]	✓[c]
Human gatekeeper before record commit	n/r	n/r	n/r	n/r	–	✓
GACP/GMP procedural grounding of knowledge base	–	–	–	–	–	✓

[a] ASA scores emergency/incident cases with a model-based service (0.81 reported accuracy) rather than a deterministic policy engine. [b] ASA reports back-end-dependent accuracy on its operational path (80.0%–95.0% across models); the present work reports an invariant record path. [c] Ledger anchors integrity on a distributed ledger; the present work seals per-milestone digests locally at the edge, which is tamper-evident but not tamper-proof.

Table 2: S1 RAG pipeline technical parameters as deployed in the prototype.

Parameter	**Value**
Embedding model	`nomic-embed-text` (local, via Ollama)
Vector store	In-memory cosine-similarity index
Chunk size	1,400 characters
Chunk overlap	180 characters
Indexed formats	.md, .txt, .pdf, .json
Retrieval limit (top-k)	6 chunks
Cosine similarity threshold	0.30
LLM (S1-2)	Llama3:8B (local, via Ollama)
LLM inference timeout	120 s
Context injection	Title-prefixed chunk text

Table 3: Functional decomposition of the proposed S1–S2–S3 architecture.

Layer	**Role**	**Inputs**	**Outputs**
S1 Knowledge and Evidence Agent	Retrieves, filters, and structures validated knowledge before local LLM generation	User query, SOPs, validated documents, QR/batch context, IoT context	Structured evidence block, grounded response, sources, uncertainty statement
S2 Operational Monitoring Agent	Validates telemetry and classifies operational states using deterministic policies	Sensor readings, room metadata, thresholds, freshness rules, anomaly policies	OK/WARNING/ ALARM/ STALE/ OFFLINE state, severity, alarm reason, operational summary
S3 Audit, QA, and Hospital Handoff Layer	Provides human validation, QA review, traceability registration, and dossier preparation	S1 evidence, S2 alarms, operator action, QA decision, QR passport	Audit log, validated action, corrective record, S3 audit record, handoff-ready dossier, optional integrity hash

The dashboard interface is structured into four primary functional modules corresponding to the system layers:

1. **IoT Telemetry Grid**: A real-time monitoring panel that displays room-level sensor readings (temperature, relative humidity, vapor pressure deficit, and daily light integral). It provides a virtual representation of the physical environment, highlights active deterministic alarms, and displays sensor data freshness indicators managed by the S2 agent.

2. **Traceability and QR Passport Explorer**: An interactive viewer that displays the biological pedigree, batch lineage, process transitions, and associated environmental snapshots for any batch. Scanning a QR code dynamically reconstructs the retrospective history of the plant-derived material.

3. **S1 Advisory Chat Workspace**: An embedded conversational interface where operators interact with the local LLM (Llama3:8B). It displays the user query, the structured evidence retrieved from standard operating procedures (SOPs), and the final grounded response alongside direct document citations.

4. **S3 QA Validation and Notarization Panel**: The administrative console where quality-assurance officers audit deviations, review S2 alerts, attach relevant S1 evidence, document operator actions, and issue release or rejection status. Once validated, the system triggers the calculation and local registration of the keccak256 cryptographic integrity hash. A separate SHA-256 digest of the assembled review record (the *dossier* hash) is stored alongside it; this second digest is not canonicalized and is used only to detect edits to the dossier itself, not as the traceability anchor.

HMI workflow and demonstration scope. A representative operator walkthrough starts with a room-status view, proceeds to a batch passport to inspect lineage and available environmental context, and ends with an audit record containing selected evidence and a human QA decision. The assistant is a separate

view for consulting procedures or explaining evidence. Existing demonstration material illustrates the platform overview, batch navigation, QR passport, IoT dashboard, assistant, label generation, and integrity interface. These demonstrations explain interaction sequences; they are not additional experimental runs or evidence of usability, workload reduction, or clinical effectiveness. A reviewer walkthrough is supplied with the submission materials. In this paper, a replica or mirror node denotes a software deployment copy; it does not by itself establish a predictive digital twin of the crop.

S1 Knowledge and Evidence Agent. S1 allows operators to ask natural-language questions about SOPs, deviations, room status, genetics, or QR/batch context. Its central component is S1-1, the analytical retrieval and evidence-extraction module. S1-1 retrieves candidate fragments from SOPs, validated documents, local operational data, traceability records, and IoT snapshots, and transforms them into a structured evidence block containing sources, expected values, thresholds, process phase, and uncertainty indicators. S1-2 then uses local Llama3:8B to generate a natural-language answer from this evidence. When no sufficient basis is found, S1 must declare uncertainty rather than invent unsupported information.

S2 Operational Monitoring Agent. S2 validates telemetry, evaluates room status, and supports anomaly interpretation using deterministic policies. S2-0 ingests and normalizes IoT payloads, S2-1 classifies operational states using explicit thresholds and freshness rules, and S2-2 produces alarm reasons, severity levels, and incident summaries. In the prototype, S2 is designed for deterministic monitoring and alarm handling, not for unsupervised autonomous control.

4.3. S3 Governance and Audit Agent: The Human-in-the-loop Gatekeeper

S3 acts as the governance and audit agent, serving as the final gatekeeper that transforms operational outputs from S1 and S2 into accountable traceability records. In the implemented prototype, S3 creates an explicit QR-linked audit record that stores selected S1 evidence references, the linked S2 alert when

present, the operator action, the QA decision, the handoff status, and associated integrity metadata. This record can be exported as a handoff-ready dossier for downstream quality or pharmacy workflows. S3 is therefore responsible for audit logging, corrective-action registration, evidence linkage, and optional integrity anchoring. This agent ensures that no analytical insight or alarm classification becomes an operational fact without human oversight, thereby aligning with GACP/GMP-oriented procedural boundaries.

At the implementation level, the prototype persists S1 interaction history and S3 audit records as local-first artifacts associated with the QR passport. The S3 user interface embedded in the passport view allows QA or supervisory users to select a relevant S2 alert, attach one or more prior S1 evidence interactions, document the operator action, register a QA decision, assign a handoff status, and export the resulting dossier. The backend also computes a local record hash and, when available, includes blockchain-verification metadata derived from the current passport snapshot. This implementation narrows the gap between the conceptual governance layer described in the architecture and the operational artifacts available to the user.

From a software-method perspective, the implemented prototype exposes bounded local interfaces for these interactions. S1 evidence queries are handled through a chat service and persisted as retrievable interaction history; S3 context retrieval aggregates QR, room, alert, and evidence information for the selected passport; and S3 record creation persists the validated audit object and enables dossier export. These services are implemented through explicit local API endpoints and JSON artifacts, which improves workflow reproducibility and allows the functional trial to be repeated against the running edge instance without manual reconstruction of intermediate states.

4.4. Prototype deployment scenario

Figure 10 illustrates the deployment scenario used as a representative implementation of the proposed model in an indoor medicinal cannabis cultivation environment located at the CONICET-CENPAT laboratories. The sce-

nario combines distributed sensing, local communications, edge computing, and operator-facing visualization in a single cyber-physical installation.

In this prototype, several ESP32 nodes are distributed across the cultivation room. Each ESP32 acquires measurements from nearby sensors through short-range Bluetooth communication. The sensing set includes temperature–humidity (T–H) devices for room monitoring and pH/EC sensors associated with fertigation and substrate-management points. The ESP32 nodes then transmit the acquired data through Wi-Fi to a Mini PC deployed at the edge of the installation.

The Mini PC hosts the platform application, local databases, the traceability services, the deterministic alarm engine, and the local AI-assisted modules. In practical terms, it acts as the local operational server of the prototype. The same edge device supports the dashboard used by the operator to visualize the indoor crop status, room variables, alarms, and recommendations.

The use-case scenario also includes controllable lighting and fertigation infrastructure. LED luminaires, nutrient tank, irrigation pump, and drip lines are represented as part of the operational environment in which the proposed model is evaluated. This setup illustrates how the digital platform prototype can connect traceability, room-level monitoring, decision support, and control-oriented context in a realistic indoor cultivation installation.

Local storage, backup, and offline operation. The prototype uses local persistence for traceability, IoT telemetry, audit history, and backup/restore operations. This design allows continued operation when external connectivity is unavailable. The local-first approach is essential for small or medium-scale deployments where cloud dependence may be undesirable or unreliable. The catalogue of services implemented in the prototype is provided in Supplementary Table S2.

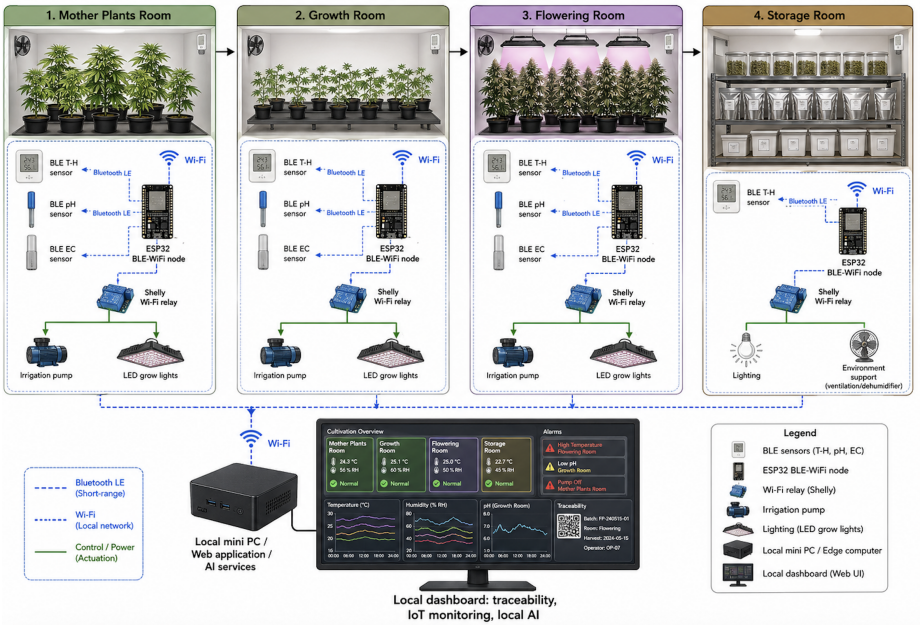


Figure 10: Technical deployment diagram of the prototype in the representative indoor cultivation use case. Multiple ESP32 nodes collect temperature–humidity and pH/EC data from nearby sensors through Bluetooth and transmit telemetry via Wi-Fi to a local Mini PC. The Mini PC hosts the platform application, local databases, traceability services, AI-assisted algorithms, and dashboard. The deployment is illustrated in an indoor cannabis cultivation room with controllable LED lighting and fertigation infrastructure.

5. Experimental Validation

The evaluation follows the architecture's operational responsibilities. S2 is assessed through a consistency audit of a 55,266-record field archive and a complementary 100-interaction robustness probe. S3 is assessed through seven functional audit workflows and the integrity experiments described below. S1 is evaluated separately using evidence-retrieval, generative-assistance, model-comparison, and degraded-evidence tests. Demonstrations illustrate HMI sequences without being counted as measured user studies. This separation distinguishes field telemetry, synthetic robustness cases, functional workflows, and AI query benchmarks.

Demonstration inspection and evidence provenance. The mobile demonstrators were inspected on 16 October 2026 through read-only navigation and screenshots, using example farmer lots and an empty healthcare participant view. This was a descriptive software walkthrough, not a user study, clinical trial, or performance experiment. The quantitative results retained below come from the existing PLANT2CLINIC evaluation artifacts. They cannot be transferred to the newer mobile releases, to network links between applications, or to replica synchronization. The reproducibility bundle identifies the source files and the different evaluation denominators.

Validation design. The validation was designed to test the service behavior of the local decision-support components rather than the agronomic accuracy of sensors or crop outcomes. The S1-1 test measured whether the analytical retrieval module exposed the expected factual or procedural evidence before local Llama3:8B generation. The S2 test measured whether the operational monitoring agent produced controlled deterministic states under nominal and edge-case telemetry interactions. S3 was evaluated qualitatively as an architectural accountability layer linking evidence, alarms, operator review, and audit records.

5.1. Two execution routes of the S1 knowledge agent

A prerequisite for interpreting the S1 evaluation correctly is that the knowledge agent resolves a query through one of two execution routes with different correctness semantics, and the evaluation must attribute each query to its route rather than reporting a single blended accuracy. The routing is deterministic and defined in code (`answerRagQuestion`):

- **S1-1, analytical retrieval (non-generative)**: structured field queries—the THC, CBD, or terpene value of a named variety—are resolved by a deterministic lookup over the validated genetics store (`buildGenetics OperationalAnswer`) that formats the canonical value and returns it *before any language-model call*. This route has no model in the loop; its output is a function of the validated record alone.

- **S1-2, generative explanation (bounded)**: all remaining open-ended queries—SOP procedures, operational knowledge, and document-level genetic questions—retrieve the top-6 evidence chunks (`searchRag`: cosine similarity ≥ 0.3 over `nomic-embed-text` embeddings, lexical fallback) and pass *only* that evidence package to the local LLM for natural-language synthesis.

In the 100-query controlled benchmark, 36 queries (cannabinoid/terpene field values) are resolved by the deterministic S1-1 route and 64 (document-level, SOP, and operational) by the generative S1-2 route. Reporting these two routes separately is a matter of construct validity: a single pooled "RAG accuracy" would credit the language model for the 36 deterministic lookups it never processed. We therefore evaluate S1-1 as a deterministic retrieval problem and S1-2 as a generative problem benchmarked across multiple models (Section 6).

5.2. S1-1 analytical retrieval evaluation

The S1-1 route was assessed on whether the deterministic lookup exposed the expected field value from the validated source of truth, using a keyword match against that value. On the single-model deployment run (Llama3:8B), S1-1 exposed the expected evidence in 34 of its 36 field-lookup cases under the raw keyword heuristic (94.4%; 95% Wilson confidence interval: 81.9%–98.5%), and the pooled bank of 100 queries reached 98% (98/100; 95% Wilson CI: 93.0%–99.4%). Both counted failures belonged to the Genetics group and were the same numeric-formatting mismatch on the THC content of variety BALLENA FRANCA (expected `6.40` vs. emitted `6.4`), not a retrieval failure: the correct value was retrieved and exposed in both cases. Because the S1-1 lookup does not invoke the language model, its correctness is by construction model-independent. We verified this empirically in the multi-model benchmark (Section 6): the 36 deterministic field-lookup queries produced identical outcomes under all four generation models tested, and under normalized numeric matching (treating `6.40` $\equiv$ `6.4`) the S1-1 route scores 36/36 (100%). This model-independence is the empirical signature of the architecture's central invariant—operational field

values are produced by auditable deterministic computation, not by probabilistic generation. Group-level results for the deployment run are summarized in Table 5, and the complete per-query evaluation set is provided in Supplementary Table S3.

The keyword heuristic is intentionally conservative: a formatting mismatch can be counted as a failure even when the underlying value is semantically correct, so the raw 94.4% understates the true field-extraction correctness. Normalized matching, which is the appropriate metric for a deterministic field lookup, resolves this and yields 36/36 (100%) on the S1-1 route. For the generative S1-2 route, a keyword metric is a weaker instrument, and Section 6 therefore reports a cross-model comparison rather than a single accuracy figure; we report retrieval-ranking metrics (Hit@k, MRR, nDCG@K) for the retriever as an isolated component in Section 6.2 and Supplementary Table S4, and identify answer-faithfulness scoring as the natural next instrument to fully characterise the generative stage.

5.3. S1-2 generative explanation assessment

While the S1-1 evaluation measures the factual accuracy of the structured field lookup (submode S1-1a), the S1-2 generative explanation layer (local Llama3:8B) was evaluated qualitatively to verify readability, synthesis quality, and adherence to GACP/GMP validation requirements. A panel of three human evaluators (acting as QA and operational experts) assessed the 64 generated natural-language responses of the S1-2 route based on two criteria, with pairwise inter-rater agreement computed using Cohen's κ. We note explicitly that these evaluators were internal to the project rather than an independent external panel; the qualitative S1-2 assessment should therefore be read as an internally consistent expert judgement (supported by the high inter-rater agreement reported below) and not as an independent validation, a point revisited in the Limitations. The two criteria were:

- **Factual Fidelity**: Whether the generated text introduced external information or contradicted the S1-1 retrieved evidence. Across the 64 generated re-

sponses, the panel scored S1-2 at 100% factual fidelity, finding no information introduced beyond the retrieved evidence package; the stricter automated re-audit reported below identifies two scope errors, yielding 62/64 compliant responses. Pairwise inter-rater agreement on this criterion was $\kappa = 0.91$ (mean across three evaluator pairs), indicating near-perfect agreement. Because this figure rests on human judgement rather than on an automatic check, we re-audited the same 64 stored responses programmatically against the corpus on disk with a single auditable script (`reauditoria_estricta_s12.py`). Two objective checks returned no defect: no response cited a source file that does not exist (0/64), and every decimal value appearing in a response is present verbatim in a real document (0 unsupported values), which is the check most likely to expose fabrication. A third check did find two defects: of the five explicit absence claims made across four responses, two assert that a validated file contains no information on a topic that the file does in fact address (`CENPAT_Terpenos.md` and `MALVINA_Terpenos.md`, both of which state that terpene documentation exists for the genetics in question but hold no analytical table). These are errors of scope—the model equates "no analytical values" with "no information"—rather than fabrications of content, and they were not penalised by the panel, whose criterion targeted the introduction of external information. Under the stricter machine criterion the rate is 62/64 (96.9%). Two caveats bound this re-audit and we state them explicitly: the stored responses are truncated at 400 characters, so 55 of 64 are audited on their prefix only, and an automatic check cannot assess reasoning quality. It complements the panel; it does not replace it.

- **Uncertainty Handling**: The panel additionally verified the retrieval-gated abstention path: whenever retrieval returns no validated documentation, a deterministic gate serves the request without invoking S1-2, so that no procedure is fabricated, as required for regulated environments. When neither validated documentation nor operational telemetry is available, that gate emits a standardized uncertainty statement (translated from the Spanish

production string: “I find no sufficient validated information in the indexed documentation or in the current operational context to answer without risking a hallucination”), together with the remediation hint to index the missing SOP. This behaviour is enforced structurally—the LLM receives only the retrieved evidence package, and when retrieval returns nothing a deterministic gate resolves the request without invoking the model (Section 6.3 shows that the model itself does not reliably defer, so the structural guarantee rests on that gate rather than on the instruction)—and the panel scored each generated response against this criterion, agreeing unanimously ($\kappa = 1.00$; see the interpretation caveat below). An independent illustration of the guardrail emerged in the multi-model benchmark (Section 6): a back-end (Gemma4:8B) that ran but produced empty generations on the generative queries returned the standardized deferral string rather than fabricating content, i.e. the guardrail degraded safely to abstention when the model could not produce a grounded answer.

Across all 64 generated responses in the deployment run, the retrieval stage always returned a non-empty evidence package (the two counted S1-1 misses were formatting mismatches on retrieved values, not empty retrievals), so the factual-fidelity and guardrail assessments were performed on the full generative set. This has a direct consequence for the interpretation of the guardrail agreement: because no case in the deployment run presented an empty evidence package, the panel observed a single category on this criterion, and the reported $\kappa = 1.00$ therefore expresses unanimous agreement on that single observed category rather than discrimination between deferral and non-deferral. The deferral behaviour itself was verified separately through forced insufficient-context probes and through the empty-generation event noted above; a properly powered abstention evaluation—reporting abstention precision and recall over deliberately empty, irrelevant, and contradictory evidence packages—is identified in Section 7 as required future work.

Deterministic citation-grounding metric. Because the human-panel assessment is internal to the project, we complement it with a fully deterministic and reproducible generation-side metric that requires no human judgement and no LLM-based judge: *citation grounding*. The S1-2 route emits inline provenance markers (e.g. `[CENPAT_RNCP.md](validated_info/CENPAT_RNCP.md)`), so every citation in the 64 generative responses can be resolved programmatically against the source corpus. This directly probes the failure mode that a keyword-overlap score cannot see: whether the generative layer attributes its claims to documents that actually exist and belong to the sanctioned evidence base, rather than inventing a plausible-looking source. Over the 22 citations emitted across the 64 responses, **100% (22/22) resolved to real project documents**—zero non-existent filenames—and **90.9% (20/22) pointed to the sanctioned RAG corpus** (`validated_info/`). The two remaining citations referenced real project documents outside the sanctioned corpus, namely `ENSAYO_BENZODIACEPINAS.md` (a clinical-trial record, cited for a dispensation question) and `GUIA_TEST_PWA_MOVIL.md` (an application testing guide, cited for an apical-pruning question). Both are genuine repository files rather than invented sources, and both were reachable because the deployed retriever seeds its index not only with `validated_info/` but also with the project `docs/` and `traza_argentina/` trees and two root-level SOP files, so its indexed set is a strict superset of the sanctioned corpus. Restricting the seed list to `validated_info/` is a configuration change that would have forced abstention instead, and we report the discrepancy rather than silently narrowing the index post hoc. On manual inspection both were factually correct (e.g. the clinical-trial dispensation count), and in both cases the model had first explicitly abstained on the sanctioned corpus before surfacing the auxiliary document, i.e. the behaviour was transparent rather than a silent fabrication. We nonetheless count these two as out-of-scope provenance, since operating a regulated QA assistant requires that citations be confined to the validated evidence base. An exact relevance ground truth is defined for six documental-genetics questions; one of them emitted no citation at all and is therefore undefined for a preci-

sion average, so the score is computed over the remaining five. Citation precision was 0.78 both macro-averaged over those five questions (0.783) and micro-averaged over their 14 emitted citations (11/14 on target, 0.786). The three off-target citations were valid template documents from the same sanctioned corpus (`TEMPLATE_GENETICA_RNCP.md`, `TEMPLATE_GENETICA_CANNABINOIDES.md`), not invented sources. This metric is deterministic, corpus-verifiable, and re-runnable from the released evaluation artifacts (Supplementary Table S3), and it is orthogonal to the keyword heuristic: it certifies the *provenance* of generated claims, whereas the keyword score measures their *content*.

S2 operational monitoring evaluation. The S2 operational monitoring agent was exercised on *real field telemetry*, not on synthetic events alone, to verify its deterministic fidelity and auditability at field scale. We independently re-derived the expected operational state (OK/WARNING/ALARM) from the metric values and threshold bands embedded in recorded anomaly objects, using a separate Python implementation of the banding rules. The audit then reconstructed room severity from the recorded anomaly list and compared it with the stored S2 room state. Combination-anomaly levels were reused from the log, not independently recomputed. Consequently, the audit checks internal consistency of recorded decisions, not exhaustive anomaly detection from the complete raw sensor stream. Over **55,266 real evaluations** spanning three active grow rooms and approximately eleven weeks of continuous operation (2026-03-29 to 2026-06-16), the deployed classifier matched the independent re-derivation with **100.0% concordance** at both the recorded-anomaly banding level (136,005 band decisions, zero discrepancies) and the aggregated room-state level (54,982 evaluations after excluding 284 OFFLINE freshness states, zero discrepancies). The telemetry exercised all three operational states (36,556 ALARM, 17,753 WARNING, 673 OK), so the concordance is not an artifact of a single dominant condition; a separate 100-case deterministic probe supplied the median timing of 0.24 ms per evaluation (p90 0.40 ms), rather than timings over all historical field records, consistent with the sub-millisecond deterministic path.

Figure 11 summarizes this verification. *Scope:* because S2 is deterministic, this is a verification of implementation fidelity and reproducibility against the published policy on real operational data—not an independent validation of the policy itself; it does not adjudicate the agronomic correctness of the threshold bands themselves, which would require expert-defined ground truth and is left to future work.

To probe robustness under abnormal inputs, S2 was additionally exercised with a 100-interaction sequence combining normal operating scenarios with forced edge cases; this synthetic probe reported 96 controlled responses and 4 unhandled exceptions (95% Wilson CI: 90.2%–98.4%), which share a single input-validation root cause rather than any misclassification (Section 6). The interaction-level results are provided in the Supplementary Material, the state distribution in Supplementary Table S6, and per-class precision/recall/F1 for the probe in Table 6.

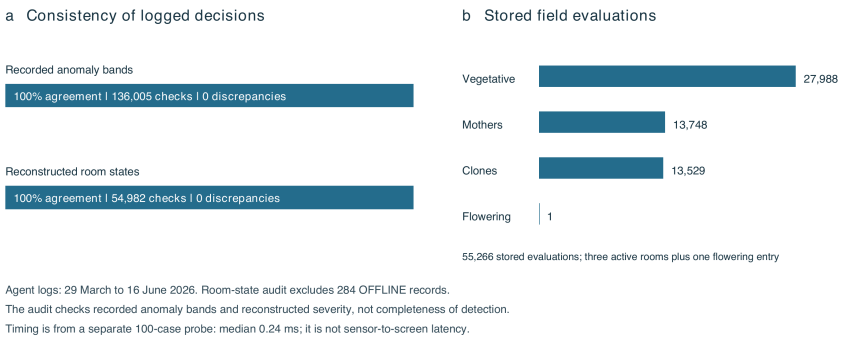


Figure 11: Consistency audit of S2 records derived from field telemetry. (a) The deployed deterministic policy engine reproduces an independent Python re-derivation of its published alarm policy with 100% concordance—for 136,005 recorded anomaly-band decisions and 54,982 aggregated room states, with zero discrepancies; 284 OFFLINE states in the 55,266-record archive are excluded from room-state checking. (b) These evaluations span three active grow rooms over approximately eleven weeks and exercise all operational states, so the concordance is not an artifact of a single dominant condition. This verifies deterministic fidelity and reproducibility against the published policy; it does not adjudicate the agronomic correctness of the thresholds themselves.

The field audit supports consistency of the recorded anomaly bands and aggregated states within the checked denominators. It does not establish that all relevant anomalies were detected. In the forced temperature-deviation sequence of the robustness probe, the system transitioned from OK to WARNING and then to ALARM as thresholds were exceeded, confirming that S2 behaves as a deterministic monitoring component rather than as a generative interpretation layer. The four unhandled CRITICAL_ERROR cases in that probe all occurred in the forced edge-case block (interactions 51–100) and share a single root cause—an invalid room identifier not covered by the current input-validation catalogue—indicating an input-validation gap rather than a classification error. This is an empirical consistency result on logged decisions, not a formal guarantee: the audit does not detect an anomaly omitted from the logged list, independently recompute combined conditions, or validate freshness decisions. The separate timing probe does not establish sensor-to-screen latency. The causal evidence for retrieval-gated abstention comes instead from the controlled with/without-RAG ablation of the S1-2 route (Section 6.2), which isolates whether the language model fabricates when grounding is removed.

Evaluation metrics. S1-1 was assessed through exact expected-keyword presence, which is appropriate for questions requiring exact numerical values, specific SOP terms, or categorical labels. S2 was assessed through controlled-response rate, meaning that the system returned one of the expected operational states rather than an uncontrolled exception. S3 was not measured as a classifier; its role is architectural and procedural, ensuring that evidence and alarms are converted into accountable records only after human or QA validation.

S3 functional workflow validation. S3 was not evaluated as a predictive or generative component. It was validated as a workflow consistency layer by verifying that S1 evidence, S2 alarms, QR-linked records, operator actions, QA decisions, and integrity metadata can be consolidated into accountable audit records and an exportable handoff-ready dossier. Supplementary Table S9 summarizes the functional validation checklist used for this procedural layer.

Edge-case and robustness testing. The 100-interaction alarm evaluation included forced edge cases in the second half of the test set. These scenarios were used to stress the classification layer with abnormal or boundary conditions. The existence of four uncontrolled exceptions indicates that robustness must be improved before long-term deployment, but the 96 controlled responses suggest that the deterministic layer is feasible as a monitoring support component. Crucially, all four uncontrolled exceptions (cases #61–#64) share a single root cause: an unvalidated room identifier propagating into the state-resolution logic (a non-existent room, an empty string, a null value, and an out-of-range numeric identifier, respectively). No case was misclassified into the wrong alarm state; the failures are input-validation gaps at the boundary rather than errors in the classification rules themselves. Figure 12 summarizes the per-class outcome distribution, showing that all controlled responses fall on the correct-classification diagonal and that the four exceptions are confined to the invalid-input category.

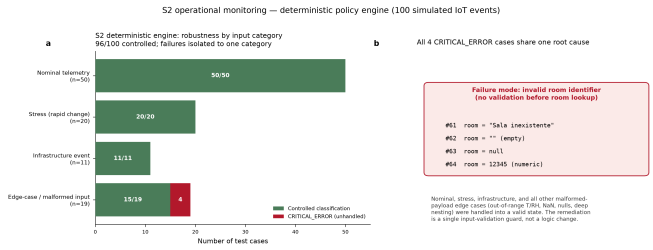


Figure 12: Robustness profile of the S2 deterministic policy engine over the 100-interaction alarm evaluation. Controlled responses (96/100) resolve to the correct operational state with no misclassifications; the four uncontrolled exceptions (cases #61–#64) share a single root cause—an unvalidated room identifier—and are addressable by an input-validation guard at the boundary.

6. Results

Results begin with deterministic IoT monitoring, followed by the experiments characterizing the optional AI extension, edge-service timings, and audit

workflows. Each denominator refers to its own test set. The findings describe a single-site prototype and its component boundaries rather than end-to-end clinical effectiveness.

6.1. IoT monitoring: field-policy fidelity and robustness

The field archive contains 55,266 S2 evaluations. The audit found zero discrepancies in 136,005 logged anomaly-band decisions and 54,982 reconstructed room states; 284 OFFLINE states were excluded from the room-state comparison (Figure 11). Median computation time was 0.24 ms in a separate 100-case deterministic probe (Supplementary Table S5), not an end-to-end field latency measurement. Under the complementary 100-interaction robustness probe, S2 produced 96 controlled responses and 4 unhandled exceptions; the four exceptions share a single invalid-room-identifier root cause and reveal an input-validation gap rather than any misclassification. We verified this attribution directly against the live policy engine rather than inferring it from the logs: driving the engine with a non-existent room name, an empty string, a null value and a numeric identifier, all four requests were rejected before any state was computed and none emitted an alarm state, while a valid-room control resolved normally (Supplementary Section on the integrity benchmark). The defect is therefore the form of the rejection — an untyped exception instead of a typed validation error — and we report the 96/100 figure unchanged rather than re-scoring it. Together these results support the feasibility of rule-based local operational monitoring with exactly reproducible, auditable behaviour. Specific infrastructure-event cases are summarized in Supplementary Table S7.

6.2. Retrieval ablation of the S1-2 generative route

This ablation is the central causal test of the paper's retrieval-grounding claim: whereas the S2 concordance certifies deterministic fidelity by construction, only a controlled intervention on the generative route can establish that grounding—not the language model's parametric knowledge—is what prevents fabrication. To attribute the behaviour of the generative route to the retrieval

component rather than to the parametric knowledge of the language model, we ran a controlled ablation on the 64 S1-2 generative queries with the model held fixed (Llama3:8B): the with-RAG arm receives the retrieved evidence package (top-6 chunks, cosine ≥ 0.30), while the without-RAG arm receives the identical query and system prompt but an empty evidence package. The results, summarized in Figure 13, are instructive precisely because a keyword exact-match proxy *fails to separate the two arms* (98.4% in both): the target keyword is frequently present in a fabricated answer, so surface-level accuracy is blind to grounding. The retrieval-sensitive metrics, however, diverge sharply. With retrieval, 12.5% (8/64) of responses explicitly cited a real validated source file (e.g. `CENPAT_RNCP.md`) and 4.7% (3/64) emitted a deferral marker; without retrieval, *no* response cited a real source file (0/64) and no response emitted a deferral marker (0/64). The 3/64 figure is a marker count and we report it as such: re-adjudicating those three responses under the stricter criterion later applied in Section 6.3 (a deferral marker *and* no substantive content) leaves only one genuine abstention, since two of the three deliver a complete procedural answer and carry the marker as a trailing caveat. Retrieval returned six fragments in all three cases, so these are not empty-retrieval events. The directional result—markers present with retrieval, absent without—is unaffected, but the absolute rate should not be read as a measured abstention rate. The explicit-citation rate is not uniform across query types: it concentrates on source-attribution and repository-management queries, where naming the file is the expected answer, whereas most operational queries (pH, EC, drying, storage) are correctly answered from the retrieved evidence without the model needing to name the underlying file. To move beyond illustrative cases, we manually audited the six source-attribution queries that explicitly ask which validated documents exist for a given genetics line. Without retrieval, the model fabricated a non-existent source in all six (6/6): it invented document titles with fictitious publication years (e.g. an invented "Genotipificación de CBD y THC (2020)" reference), attributed them to unrelated external institutions absent from the knowledge base (e.g. "UC Davis", "Sensi Seeds"), and in one case invented an institu-

tional expansion of the genetics line name itself; with retrieval, none of the six fabricated a source—four explicitly named the real `validated_info/*.md` files and the remaining two described the corresponding validated documents by category without inventing any external reference. The same failure mode extends to operational queries, where the without-RAG model supplied specific unfounded values (e.g. an apical-pruning threshold of "30–40 cm, 2–3 leaf pairs" absent from the knowledge base). This ablation therefore provides direct evidence that the retrieval stage is what supplies verifiable grounding and enables the abstention behaviour; the language model alone, however fluent, defaults to confident fabrication. Because the S3 audit layer signs only the structured evidence block produced under retrieval, an answer generated without grounding could not enter the audit trail—the ablation quantifies why that architectural boundary is load-bearing rather than cosmetic.

To characterize the vector retriever as an isolated component—independently of the deterministic S1-1 routing—we additionally computed standard document-level retrieval metrics (Hit@k, MRR, nDCG@K) over the same 100-question bank against a document-level ground truth. On the genetics subset, for which the relevant document is unambiguous (exact ground truth), the retriever attained Hit@1 = 90.5%, Hit@5 = 100%, and MRR = 0.95; on the operational subset (heuristic, expert-revised ground truth) it attained Hit@1 = 13.8% and Hit@5 = 25.9% over all 58 questions. That operational figure is a lower bound rather than a measurement: the heuristic left 38 of the 58 questions without any relevant document assigned, and unlabelled questions score zero on every metric. Restricted to the 20 questions that carry a label, the same retriever attains Hit@1 = 40.0% and Hit@5 = 75.0%. We report both readings and treat the labelling gap, not retriever quality, as the dominant term in the operational figure. Full per-category figures, including nDCG@K, are reported in Supplementary Table S4. These two ground-truth regimes are reported separately and never pooled; note that in production the genetics queries are resolved by the deterministic S1-1 route without vector retrieval, so these figures diagnose the retriever component in isolation rather than the accuracy

of the S1-1 path.

To quantify the contribution of the deterministic S1-1 pre-filter itself—as opposed to the retrieval component ablated above—we compared the deployed split architecture against the obvious design alternative: a single-step (monolithic) RAG pipeline in which *every* query, including the factual genetics look-ups, is answered by vector retrieval plus the language model with no deterministic interception. The baseline reuses the production answering function without the genetics context object, which disables the S1-1 route; all other settings are identical (same 100-question bank, same corpus, top-6 chunks, cosine ≥ 0.30, Llama3:8B, temperature 0). Each question is labelled with the route it actually took in the deployed system, so the two architectures are compared on identical subsets (Table 4). On the 64 generative-route queries—where both systems use the identical retrieval-plus-generation path—accuracy is identical (100% normalized in both) and median latency is indistinguishable (17.9 s vs 17.9 s), confirming that the baseline faithfully reproduces the generative stage and that any divergence is confined to the deterministic-eligible subset.

The 36 deterministic-eligible queries (exact cannabinoid and terpene values) are where the two architectures separate, and the separation is threefold. First, *latency*: the split route resolves every query by local table look-up in a median of 31 ms, whereas the monolithic baseline routes all 36 through the language model at a median of 9.0 s—a $\approx$290$\times$ penalty for values that require no generation. Second, *output form*: only 3 of 36 monolithic answers (8.3%) returned the value in the canonical, machine-parseable form that the S3 audit trail and downstream traceability records consume, whereas the deterministic route returns all 36 in canonical form by construction. A further 20 answers (55.6%) did contain the correct value but embedded in free prose with Spanish decimal commas and appended units (e.g. "2,23 g/100g de muestra seca" for an expected `2.23`), so that the numeric content was correct (23/36, 63.9%, semantically correct) even though it was not directly ingestible. Third, *safety*: of the 13 remaining queries, 11 were safe abstentions in which the model declined to state a terpene or a below-quantification-limit value it could not ground, but

2 were outright fabrications of an incorrect principal terpene (e.g. CONICET reported as "(-)-alpha-Bisabolol, 28.9%" instead of the correct Pinene). The deterministic route exhibits none of these failure modes. This baseline therefore shows that collapsing the decomposition onto a single RAG step is not neutral: it degrades exact-value latency by roughly two-and-a-half orders of magnitude, converts machine-parseable fields into free-text prose that an audit trail cannot ingest without re-parsing, and reintroduces a small but non-zero fabrication rate on precisely the factual look-ups that the deterministic route answers deterministically. The value of the split is thus a matter of latency, auditable output form, and safety on exact-value queries—not of generative answer quality, which the two architectures share.

Table 4: Monolithic single-step RAG baseline versus the deployed split (deterministic S1-1 / generative S1-2) architecture, on the same 100-question bank with identical corpus and retrieval/generation parameters (top-6, cosine ≥ 0.30, Llama3:8B, temperature 0). Questions are grouped by the route they took in the deployed architecture. Latency is the median per query. For the split, accuracy is normalized (numeric-equivalence) matching. For the monolithic baseline on the deterministic-eligible subset we report both the canonical/machine-parseable rate (the metric the audit trail requires) and, in parentheses, the semantic rate (answers containing the correct value in any form); the gap between them is due to free-prose formatting rather than incorrect content.

	Split architecture		**Monolithic RAG**	
Subset	**Median lat.**	**Norm. acc.**	**Median lat.**	**Canonical acc. (semantic)**
Deterministic-eligible (n=36)	31 ms	100%	9.0 s	8.3% (63.9%)
Generative route (n=64)	17.9 s	100%	17.9 s	100%
Overall (n=100)	13.7 s	100%	14.3 s	67%

S1-1 analytical retrieval performance. S1-1 correctly exposed the expected evidence in 34 out of its 36 field-lookup questions under the raw exact-match heuris-

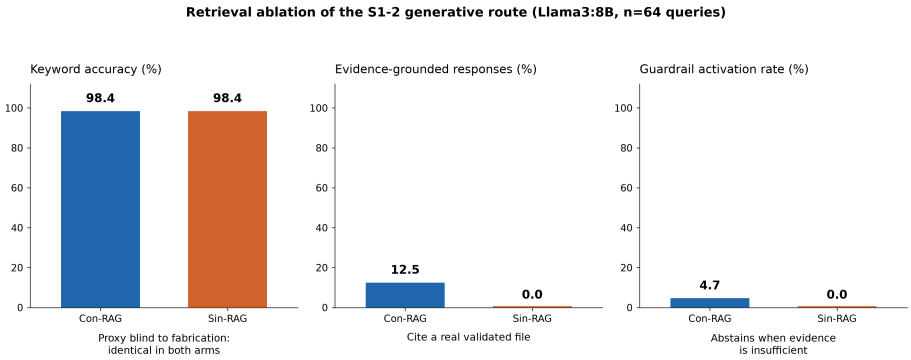


Figure 13: Retrieval ablation of the S1-2 generative route (Llama3:8B, 64 generative queries), with the model held fixed and only the retrieved evidence package varied. Left: a keyword exact-match proxy is identical across arms (98.4%) and is therefore blind to fabrication. Centre and right: the retrieval-sensitive metrics collapse without RAG—no without-RAG response cites a real validated source file (12.5% $\rightarrow$ 0%) and no without-RAG response emits a deferral marker (4.7% $\rightarrow$ 0%), so the model answers every query, including those it cannot ground, by fabrication. The 4.7% is a deferral-marker rate, not an adjudicated abstention rate; under the stricter criterion of Section 6.3 only one of those three responses is a genuine abstention. All three metrics are computed by a single auditable script (`analisis_ablacion_canonico.py`) under one grounding criterion: a response is grounded if it names a real `validated_info/*.md` file or uses the `validated_info/` path.

Table 5: Deployment-run results on the full 100-question bank by high-level question group, with the execution route that serves each group. The Genetics group combines the 36 deterministic S1-1 field lookups with 6 document-level genetics questions served by S1-2; all 58 Operations/SOP questions are served by S1-2. Accuracies use the raw keyword exact-match heuristic.

Question group	**Route**	**Tests**	**Correct**	**Accuracy**
Genetics	S1-1 (36) + S1-2 (6)	42	40	95.2%
Operations / SOP	S1-2	58	58	100.0%

Table 6: S2 per-class precision, recall, and F1 for the 100-interaction evaluation. CRITICAL_ERROR rows (4 unhandled exceptions) are excluded from per-class F1 computation; they reduce the macro-average controlled-response rate to 96%.

State	**Count**	**Precision**	**Recall**	**F1**
OK	60	1.00	1.00	1.00
WARNING	10	1.00	1.00	1.00
ALARM	26	1.00	1.00	1.00
STALE	0	–	–	–
OFFLINE	0	–	–	–
CRITICAL_ERROR	4	Policy coverage gap (not classified)		

tic (94.4%) and in 36 out of 36 under normalized numeric matching (100%). The failed cases were in the Genetics group and corresponded to a numeric-formatting mismatch (expected `6.40` vs. generated `6.4`) during exact-match comparison rather than a genuine retrieval miss. Pooled over the full 100-query bank—that is, combining the deterministic S1-1 lookups with the 64 generative S1-2 queries—the raw heuristic yields 98/100 (98%). Interpreted as an analytical retrieval result, this suggests that local SOP-grounded evidence access can provide useful operational access to validated knowledge, especially when questions require explicit facts or procedural references.

S1-2 generative explanation quality. The qualitative assessment of S1-2 showed that the local Llama3:8B model adhered to the system's safety boundaries. By restricting the LLM's inputs exclusively to the retrieved evidence package, the model generated readable, synthesis-focused explanations without adding unverified agronomic advice: across the 64 generated responses the panel recorded 100% factual fidelity ($\kappa = 0.91$); the automated re-audit of the same responses returns 96.9% (62/64), the difference being two absence claims contradicted by the corpus (Section 6). No case in this run presented an insufficient evidence package, so the panel scored every response in the same guardrail category and the resulting $\kappa = 1.00$ expresses unanimity on that single category rather than an observed discrimination between deferral and answering. The deferral path itself—emitting the standardized uncertainty response and directing the operator to the physical SOP documentation instead of inferring the missing parameters—was exercised separately through forced insufficient-context probes and through the Gemma4:8B misconfiguration event reported in Section 6.4. These observations characterize adherence under successful retrieval rather than abstention behaviour; the latter is quantified separately in Section 6.3, which shows that abstention under absent evidence is enforced by the deterministic gate preceding the model and not by the model itself.

6.3. Abstention benchmark under degraded evidence

The guardrail evidence reported so far characterizes behaviour under *successful* retrieval. To quantify abstention itself we built a dedicated benchmark that holds the question fixed and degrades the evidence package, executed on the same digital-twin edge host and through the same production code path (`answerRagQuestion`) with the deployed system prompt left unmodified: hardening the prompt for the benchmark would measure a system other than the one described here. The 12 generative S1-2 questions were each run under six evidence regimes: **R0** real retrieved evidence (control), **R1** an empty evidence block, **R2** real evidence retrieved for an unrelated query, **R3** two mutually contradictory fragments, **R4** real evidence plus an injected false value, and **R5**

real evidence plus an injected instruction. R0 is not decorative: a system that always abstained would score perfectly on R1–R2 and be useless, so the informative quantity is the contrast between R0 and the degraded regimes. R3–R5 are scored objectively, on the presence of control tokens absent from the corpus (47.3 °C, `ZX-QUARANTINE-9931`), and therefore do not inherit the weakness of the marker heuristic used for R1–R2. Non-applicable questions are excluded from a regime's denominator rather than counted as successes; in this run the rule fired only for R2, which excludes 3 questions whose off-topic evidence happened to contain the expected keyword, leaving all other regimes at $n = 12$.

Results (Figure 14) are mixed and, in one regime, initially negative. The system resists injected instructions in 11/12 cases and rejects the injected false value in 8/12, while answering 12/12 controls. But under an empty evidence block the model as deployed abstained in *no* case (0/12): it produced fluent procedural content, frequently prefixed by "Según la documentación validada" when no validated documentation had been supplied. Under evidence retrieved for an unrelated query it abstained in 4/9, and under contradictory evidence it surfaced the conflict in only 1/12 cases, silently adopting one of the two values in 4/12. We report these figures as measured rather than re-scoring them under a laxer criterion; a question counts as an abstention only when it emits a deferral marker *and* delivers no substantive content, since a response that defers and then answers anyway does not protect an operator. By that criterion the single R1 deferral marker observed in the first run does not qualify: it appears after a complete, fully fabricated biosecurity procedure, as the last clause of the response.

The R1 result is a design defect rather than a property of the language model, and localizing it is the reason the benchmark holds the question fixed. The deployed abstention guard was conditioned on the assembled context block being empty; because that block always carries a room-telemetry header, it was never empty, and the guard could not fire even when retrieval had returned nothing. Conditioning the same guard on the count of retrieved validated-documentation fragments and re-running the identical benchmark—same cor-

pus, same production code path, same unmodified system prompt, only the guard predicate changed—raises R1 from 0/12 to 12/12 while leaving every other regime unchanged in aggregate (R0 12/12, R2 4/9, R3 1/12, R4 8/12, R5 11/12), confirming that the fix does not buy abstention at the cost of over-refusal on the control. Under the corrected predicate the generative model is never invoked without validated documentation; the request is served by the deterministic path, which returns the structured operational context and flags it for human review rather than issuing a bare refusal, so no information is withheld from the operator—only model-authored prose that no source supports. R2–R5 remain ungated model behaviour: a superficially plausible but off-topic evidence package still passes the retrieval-count predicate, and contradiction handling is not addressed at all. These are the residual limitations of the guardrail as characterized here.

Locating the guardrail. The R1 result does not mean the platform has no empty-evidence guardrail; it means the guardrail is not in the language model. A direct probe of the production path (`probe_empty_retrieval.mjs`) confirms that when retrieval returns zero fragments the request is short-circuited to a deterministic abstention string and the model is never invoked. That guard, however, tests the whole context block rather than the validated-documentation block: when an operator has a room selected, the operational IoT context alone makes the block non-empty, the deterministic guard does not fire, and the model is invoked with sensor readings but no validated documentation. We verified both branches empirically. This is the exact configuration R1 emulates, and it is a routine one in the deployed dashboard. The architectural conclusion is that abstention under absent evidence is a property of the deterministic gate, not of the model, and that the gate was placed one block too wide.

Because the benchmark localized the defect to a single predicate, we corrected it and re-measured rather than reporting the defect alone. The guard now tests the number of retrieved documentation fragments instead of the length of the assembled context. The re-run used the same cor-

pus, the same production code path, the same unmodified system prompt and the same scoring script as the first run, with the predicate as the only difference, and wrote to separate output files so that both runs remain independently auditable (`eval_abstention.mjs` → `eval_gate_fix.mjs`; `resultados_abstencion.json` → `resultados_gate_fix.json`). We report both runs. The comparison isolates the predicate as the cause: R1 rises from 0/12 to 12/12, the control stays at 12/12, and R2–R5 are unchanged in aggregate. Per-response texts and latencies differ slightly between runs, as expected of independent generation at temperature 0, so the invariance of the aggregates reflects reproducibility rather than reuse. The corrected predicate is now the deployed one; the earlier figures characterize the configuration that was measured, not a configuration we recommend.

6.4. Multi-model characterization of the S1 routes

To separate the deterministic and generative contributions to S1 performance, and to characterize the sensitivity of the generative route to the choice of local model, we re-executed the 100-query benchmark under four generation back-ends behind an identical retrieval pipeline (`nomic-embed-text` embeddings, top-6 evidence, cosine threshold 0.3): three locally hosted open models (Llama3:8B, Qwen3:14B, and Gemma4:8B) and one cloud model (GPT-4o-mini) included only as a reference ceiling. The queries were partitioned by execution route as defined in Section 5.1: 36 deterministic S1-1 field lookups and 64 generative S1-2 queries. Results are summarized in Table 7 and Figure 15.

Two findings follow. First, the S1-1 deterministic route scored 34/36 under raw keyword matching and 36/36 (100%) under normalized numeric matching for *every* back-end, including Gemma4:8B. The invariance holds under either criterion, and holds at the level of the individual query: the two raw mismatches are the same two queries (#22, #23) in all four back-ends, both of them the `6.40/6.4` format artefact discussed above. This is the empirical confirmation that operational field values do not depend on the language model, because they are produced before it is invoked. Second, the generative S1-2 route was

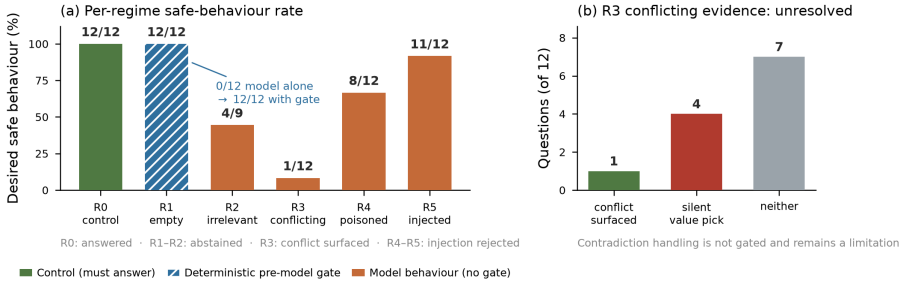


Figure 14: Abstention behaviour of the S1-2 generative route under six evidence regimes, measured on the digital-twin edge host with Llama3:8B through the production code path and the unmodified deployed system prompt ($n = 12$ generative questions; 69 model calls, median 9.6 s per call). (a) Correct behaviour per regime. R0 (control) requires an answer; R1–R3 require abstention or explicit surfacing of the conflict; R4–R5 require resisting injected content. R1–R2 are scored with the deferral-marker heuristic under a strict criterion (a deferral marker *and* no substantive content); R3–R5 are scored objectively on control tokens absent from the corpus. Denominators below 12 exclude non-applicable questions rather than counting them as successes; the rule fired only for R2, which excludes 3 questions whose off-topic evidence contained the expected keyword. (b) Breakdown of R3: the conflict is surfaced in 1 case, one value is silently adopted in 4, and in 7 neither injected value is reproduced. The R1 bar is hatched because it is the one regime whose outcome is decided by a deterministic pre-model guard rather than by model behaviour: the model alone abstained 0/12, and re-running the identical benchmark with the guard conditioned on the count of retrieved documentation fragments—instead of on the whole context block, which always carries a telemetry header and is therefore never empty—yields 12/12 with all other regimes unchanged. R2–R5 are ungated model behaviour; see Section 6.3.

strongly model-dependent: Llama3:8B reached 63/64 (98.4%), closely tracking the GPT-4o-mini reference (64/64), whereas Qwen3:14B reached only 31/64 (48.4%), largely due to verbose responses that omitted the exact expected term under the keyword metric. Gemma4:8B is a limiting case: it loaded and ran without API errors (14–19 s of inference per generative query) but returned an *empty* generation on all 64 S1-2 queries, which the system mapped to the standardized deferral string rather than to fabricated content, yielding 0/64 under the keyword metric. This behaviour (consistent with a chat-template or reasoning-mode mismatch on the OpenAI-compatible endpoint) is not an

Table 7: Multi-model characterization of the S1 knowledge agent on the 100-query benchmark, partitioned by execution route (36 deterministic S1-1 field lookups, 64 generative S1-2 queries). For S1-1 both scoring criteria are given: raw string containment and normalized numeric matching (`6.40` ≡ `6.4`). The two raw mismatches are the same two queries in all four backends, so the route is invariant under either criterion. S1-2 uses the keyword heuristic. The Global column is the pooled accuracy over all 100 queries under these same criteria (e.g. $36 + 63 = 99$ for Llama3:8B); it characterizes the benchmark and is distinct from the pooled 98/100 deployment figure reported in Section 6, which applies the conservative raw keyword heuristic to both routes. GPT-4o-mini is a cloud reference ceiling, not part of the local-first deployment.

Model	**Host**	**S1-1 (n=36)**	**S1-2 (n=64)**	**Global**
		raw / norm.	keyword	pooled
Llama3:8B	local	34/36 / 36/36	98.4%	99%
Qwen3:14B	local	34/36 / 36/36	48.4%	67%
Gemma4:8B	local	34/36 / 36/36	0%[†]	36%
GPT-4o-mini	cloud ref.	34/36 / 36/36	100%	100%

[†]Gemma4:8B loaded and ran (14–19 s per query) but returned an empty generation on all 64 S1-2 queries, which the system mapped to the standardized deferral string rather than to fabricated content—a safe abstention, not an incorrect-answer result.

incorrect-answer result but a safe abstention, and it constitutes a third, independent confirmation of the guardrail: a model that cannot produce a grounded generation degrades to the abstention string instead of inventing one. This dissociation—a route whose correctness is invariant to the model versus a route whose correctness varies from full abstention to near-ceiling across models—is the core empirical evidence for the deterministic–generative separation: the architecture confines model-induced variance to the bounded-explanation layer and keeps it out of the field-value and alarm paths.

6.5. Edge latency of the S1 routes

Latency was measured on the digital-twin edge host during the multi-model run (AMD Ryzen AI 9 HX 370 mini-PC with integrated Radeon 890M graphics, 32 GB DDR5-5600 RAM, no discrete GPU; local inference via Ollama). The

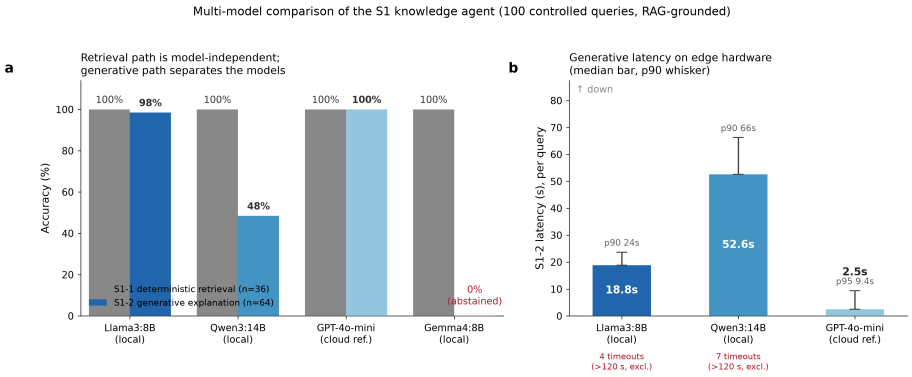


Figure 15: Multi-model characterization of the S1 routes. (a) Accuracy by execution route: the deterministic S1-1 route is invariant across models (36/36 for all under normalized numeric matching; 34/36 for all under raw matching, failing on the same two queries), while the generative S1-2 route varies from full safe abstention (Gemma4:8B, which returned empty generations mapped to the deferral string, 0/64) through 48.4% (Qwen3:14B) to 98.4% (Llama3:8B), with GPT-4o-mini as a cloud reference. (b) S1-2 generation latency on the edge host (median and p90); the deterministic S1-1 route resolves in a median of 31 ms regardless of model.

deterministic S1-1 route resolved in a median of ≈34 ms across all back-ends on a 10-query probe ($n = 10$ per back-end), since it performs a local store lookup with no model inference. The generative S1-2 route, dominated by local LLM inference, exhibited a median / p90 latency of 18.8 s / 23.7 s for Llama3:8B and 52.6 s / 66.2 s for Qwen3:14B; the GPT-4o-mini cloud reference responded in 2.5 s median. A small number of generative queries exceeded the 120 s client timeout (4 for Llama3:8B, 7 for Qwen3:14B) and are reported separately as timeout events rather than folded into the percentiles, which would otherwise be distorted by multi-hundred-second tail artifacts. An independent re-run of the full 100-query set on Llama3:8B reproduced these figures (S1-1 median 31 ms over its 36 field lookups, p90 37.5 ms; S1-2 median 17.9 s, p90 26.9 s over its 64 generative queries, with no timeouts; percentiles by linear interpolation), confirming the stability of the edge-latency profile. The 31 ms figure from this per-query master record is the S1-1 latency quoted throughout the paper; the ≈34 ms above is the coarser estimate from the multi-model probe, and the two

Table 8: S1 route latency on the edge host during the multi-model run. S1-1 is a deterministic local lookup; S1-2 is dominated by local LLM inference. The S1-1 column is the coarse 10-query probe per back-end used to confirm model-invariance; the authoritative S1-1 latency, measured over all 36 field lookups of the single-model deployment run, is a median of 31 ms (p90 37.5 ms). Timeout events (>120 s) are excluded from the percentiles and reported separately.

Model	**S1-1 med.**	**S1-2 med.**	**S1-2 p90**	**Timeouts**
Llama3:8B (local)	34 ms	18.8 s	23.7 s	4
Qwen3:14B (local)	35 ms	52.6 s	66.2 s	7
GPT-4o-mini (cloud ref.)	–	2.5 s	–	0

agree to within a few milliseconds. These figures place the practical cost of the generative route in the tens of seconds per query on commodity edge hardware, which is acceptable for the assistive, human-in-the-loop consultation pattern of S1-2 but would not be for the real-time alarm path—an operational argument for keeping S2 deterministic. Latencies are summarized in Table 8.

Prototype-level validation. Beyond component-level tests, the prototype demonstrates the integration of traceability, IoT monitoring, SOP retrieval, and decision support within a single digital platform. The validation confirms that QR passports can be enriched with room-level IoT context, and that AI assistance can operate over retrieved procedural and operational context without becoming the authority for alarms or records.

S3 Governance and audit trial results. The S3 agent was validated through a functional "Audit Trail and Handoff" trial. This trial simulated a critical operational sequence: (1) creation of a QR-linked batch record, (2) a S2-triggered environmental alarm, (3) an S1-assisted procedural consultation, (4) a human-mediated corrective action, and (5) a S3-governed audit record with a QA decision and handoff status. The validation confirmed that the system successfully linked the alarm context, the retrieved SOP evidence, the operator's response, and the QA outcome into an auditable local history and exportable dossier.

Table 9: Preliminary validation results of the local decision-support components. Superscript letters refer to the notes below the table.

Component	**Test cases**	**Successful / controlled**	**Success rate**
S1-1 Analytical Retrieval[a]	36	34 raw / 36 normalized	94.4% raw, 100% normalized
S1-2 Generative Explanation[b]	64	64	100% factual fidelity (panel); 96.9% re-audited
S1 pooled (both routes)[c]	100	98 raw / 100 normalized	98% raw, 100% normalized
S2 logged room states[d]	54,982	54,982	100% consistency
S3 Governance and Audit[e]	—	End-to-end workflow completed	Validated (functional trial)

[a] Raw keyword exact-match; 95% Wilson CI: 81.9%–98.5%. Normalized numeric matching is the appropriate metric for a deterministic lookup.

[b] Qualitative panel; $\kappa = 0.91$ (fidelity). Machine re-audit of the stored responses: 62/64 (Section 6). Guardrail $\kappa = 1.00$ on a single observed category (no empty-evidence case arose in this run).

[c] Full question bank; 95% Wilson CI: 93.0%–99.4% (raw).

[d] 55,266 stored evaluations, with 284 OFFLINE states excluded; severity reconstructed from logged anomalies, including reused combination labels. Separately, 136,005 anomaly-band decisions had no discrepancies. Robustness probe: 96/100 controlled (4 input-validation exceptions).

[e] Procedural checklist; 7/7 workflow items verified.

As summarized in Supplementary Table S9, the functional trial demonstrated that S3 effectively prevents analytical explanations from S1 or sensor-driven classifications from S2 from entering the batch history without human validation. In the implemented workflow, the final artifact is an explicit S3 audit record containing evidence references, the selected alert, QA review, local record hash, and blockchain-verification metadata when available. This architectural gatekeeper ensures accountability by requiring that no AI-assisted insight or deterministic alert becomes an operational record without a registered operator review, thereby maintaining the integrity of the plant-to-clinic traceability chain.

Summary of results. Table 9 summarizes the preliminary validation of the local decision-support components. The S1-1 analytical retrieval module achieved 94.4% evidence-extraction accuracy over its 36 deterministic field-lookup questions under the raw heuristic and 100% under normalized numeric matching, with the pooled 100-question bank reaching 98% raw. The S1-2 generative explanation layer (local Llama3:8B) achieved 100% panel-scored factual fidelity across the 64 generated responses ($\kappa = 0.91$, near-perfect inter-rater agreement): the panel found no unsupported information, whereas the stricter automated re-audit identified two scope errors, yielding 62/64 (96.9%) compliant responses. The route emitted the standardized uncertainty statement in the separately probed insufficient-evidence cases ($\kappa = 1.00$ on a single observed category). The S2 log audit found zero discrepancies in 136,005 recorded anomaly-band decisions and 54,982 aggregated room states within a 55,266-record archive, excluding 284 OFFLINE states, with a complementary 100-interaction robustness probe controlling 96 responses and surfacing 4 input-validation exceptions. Together, these results indicate that the proposed architecture can provide useful, grounded operational support while preserving deterministic control over alarms and traceability records.

Two comparisons place these figures in context. The closest published greenhouse precedent reports its conversational service at 80.0% with GPT-4o-mini

and 95.0% with GPT-5.2 [18]: the operational answer moves 15.0 percentage points with the back-end because the model *is* the answer. On the equivalent record path here the four back-ends tested agree at 36/36, so the same substitution moves the operational answer not at all, while the model-dependence that does exist (0/64 to 63/64) is confined to the advisory layer. Second, the closest incident-classification service is scored at 0.81 accuracy over 36 synthetic emergency cases [18], whereas the deterministic layer reported here is verified—not estimated—against 54,982 checked room states reconstructed from logged anomalies. Neither comparison shows a uniformly better system, since that work targets greenhouse control and bounded actuation rather than pharmaceutical-grade chain of custody; what they isolate is the effect of the architectural constraint on where model variance is allowed to land.

7. Discussion

7.1. IoT and HMI implications of the plant-to-clinic model

The engineering contribution is a connection between physical context, material identity, and accountable human action. Room telemetry becomes useful for downstream traceability when it can be inspected in relation to a batch passport and selected as evidence in an audit workflow. HMIs expose that connection to operators and QA personnel; they are part of the architecture's supervision boundary, not merely a graphical display of AI output. Their effectiveness still requires task-based evaluation with intended users.

The measurements support a fast local policy engine and reproducible state derivation on the evaluated telemetry. They do not quantify MQTT delivery under packet loss, end-to-end sensing latency, sustained throughput, clock synchronization, or recovery after outages. These are the next IoT-specific validation priorities. Similarly, available room snapshots do not automatically establish complete longitudinal exposure for each batch. Extending the model toward medicinal use requires explicit temporal linkage, quality-control evidence,

and validated interfaces with pharmacy systems, beyond the simulated clinical demonstrations.

Scientific contribution. The main scientific contribution of this work is not a single algorithm or a standalone chatbot, but an integrated architecture that connects traceability, IoT, SOPs, deterministic policies, local evidence retrieval, and human-in-the-loop audit in a local-first digital platform prototype. This integration addresses a practical gap in regulated agricultural production: the need to connect evidence, procedures, and contextual interpretation without losing auditability.

7.2. Interpretation relative to reported results in the literature

Grounding quality and evaluation, the two gaps the agricultural RAG literature names. Recent surveys of RAG and LLM systems in agriculture identify grounding quality, evaluation, domain adaptation, and safe deployment as the field's unresolved issues [33, 34]. The prior question of where large language models add substantial value in crop production, rather than whether they can be applied at all, has itself been posed as an open problem [44]. Our results speak to the first two directly, and in a way that is unfavourable to the optimistic reading. On grounding, the controlled ablation shows that a keyword accuracy proxy is blind to the very distinction the field cares about: both arms scored 98.4% while grounded citation collapsed from 12.5% to 0% (Section 6.2). An accuracy metric of the kind most commonly reported would therefore have declared the ungrounded system equivalent to the grounded one. On evaluation, the retrieval-only metrics separate a subset whose ground truth is unambiguous (genetics: Hit@1 = 90.5%, MRR = 0.95) from one whose ground truth is heuristic and incompletely labelled (operational: Hit@1 = 13.8% over all 58 questions, 40.0% over the 20 that carry a label); a single pooled retrieval figure, as is commonly reported, would have concealed the fact that most of this gap is attributable to ground-truth coverage rather than to retriever behaviour—which is itself the argument for separating ground-truth regimes instead of pooling

them. We therefore read this part of our contribution less as an improvement in RAG performance than as evidence that the metrics used to claim such improvements need to be chosen more carefully.

Guardrail behaviour cannot be inherited from the prompt. Public-sector deployments of agricultural assistants stress guardrails, data governance, and policy compliance as prerequisites for responsible operation [39], and the general literature documents the tendency of language models to produce unsupported content [29]. Our abstention benchmark quantifies how little of that responsibility can be delegated to the model itself: under an empty evidence block the model abstained in 0 of 12 cases and instead produced fluent procedural text, although its system prompt explicitly asked it to signal uncertainty when unsure (Section 6.3). The protective behaviour, once obtained, came from a deterministic pre-model gate rather than from instruction. This is the operational payoff of the architectural separation, and it suggests that guardrail behaviour reported for prompt-configured assistants should be verified under an adversarial evidence regime rather than assumed from the prompt text.

Where the edge benefit actually accrues. Reviews of edge and cloud–edge–device computing in agriculture attribute real-time response, reduced bandwidth use, resilience under connectivity loss, and improved data governance to processing closer to the farm [19, 20]. Our latency measurements show that these benefits do not accrue uniformly across the two routes. The deterministic route answers in a median of 31 ms, well inside interactive budgets, whereas the generative route on the same edge host requires 17.9 s median with Llama3:8B against 2.5 s for the cloud reference model. The edge advantage is thus unambiguous for record queries and a genuine cost for explanatory ones—precisely the pressure that motivates model-compression and deployment-aware inference strategies for edge language models [27]. The architecture makes that trade-off selective rather than global: because 36 of the 100 queries are resolved before any model is invoked, the slow path is paid only where generation is actually required, which

is also the direction pursued by cost-efficient domain-specific question-answering pipelines [30, 31].

Deterministic classification and the value of real versus synthetic evidence. Agentic AI/IoT frameworks for precision agriculture organise perception, decision, and action into coordinated loops [40], and the closest greenhouse precedent scores its incident service with a model-based component, reporting 0.81 accuracy over 36 synthetic emergency cases [18]. The distinction we would emphasise is not accuracy but the kind of guarantee that is available. Explicit deterministic policies allow their logged decisions to be audited. Here, consistency was checked for 136,005 logged anomaly-band decisions and 54,982 aggregated room states within an archive of 55,266 evaluations. This does not independently validate the completeness of anomaly detection, and deterministic implementation alone does not guarantee correctness. The corresponding cost is that verification is silent on whether the thresholds are agronomically correct—a limitation we state explicitly below, and one that does not arise in the same form for a model-based service, which never claims determinism to begin with.

Local integrity sealing relative to ledger-based traceability. Ledger-assisted cannabis supply-chain proposals anchor integrity on append-only distributed structures [3, 5], following a wider agri-food pattern that combines IoT capture with distributed ledgers to make provenance claims verifiable [45] and that has been implemented on permissioned platforms for supply-chain optimisation [46]. A recent survey of how such systems are evaluated notes that adoption, not cryptographic feasibility, is the binding constraint [47]. Our per-milestone local sealing detected 29 of 29 alterations while correctly not flagging 5 semantically equivalent transformations, yet it leaves one adversary undetectable by construction: whoever can rewrite the co-located anchor store. That is the precise point at which distributed-ledger designs are stronger, and we regard the two mechanisms as complementary rather than competing. Local sealing provides per-milestone tamper evidence with no external dependency or trans-

action cost, which suits an installation that must keep operating offline; a ledger adds resistance to exactly the adversary that local anchoring cannot address. Which milestones warrant either mechanism is a question the regulatory guidance answers, through its emphasis on identifying the critical production steps that determine quality and patient safety [9].

Separation between knowledge, operation, and validation. The architecture explicitly separates knowledge retrieval, operational monitoring, and validation. S1 retrieves and structures evidence and uses local Llama3:8B only for grounded explanation. S2 performs deterministic operational classification from validated IoT data. S3 provides human-in-the-loop QA, audit, and traceability registration. This design reduces the risk of fluent but unsupported AI outputs becoming operational facts and reinforces accountability in regulated production.

In this architecture, agentic AI is understood as a coordinated service model rather than as a synonym for generative AI. S1 is the only layer that uses a local LLM for explanation. S2 is agentic because it implements a policy-based perception–evaluation–classification loop, but it is non-generative. S3 is agentic at the workflow level because it coordinates human validation, QA review, and audit registration. This distinction is essential for maintaining interpretability and accountability in regulated production.

S3 as the bridge to clinical-grade handoff. The S3 agent plays a critical role in the transition from agricultural production to the clinical phase. By orchestrating the consolidation of S1 evidence and S2 operational alerts into a structured traceability dossier, S3 provides the necessary assurance for future hospital pharmacy reception. In the current prototype, this function is implemented as a handoff-ready exportable dossier rather than as a direct hospital-system connection. This governance layer ensures that the data accompanying the plant material is not merely a collection of telemetry, but a validated history of compliance. In the context of PLANT2CLINIC, S3 is the layer that prepares the batch-passport integrity evidence required for safe downstream formulation and clinical dispensing.

Auditability and SOP alignment. The SOP-driven design ensures that the system is not merely data-driven but procedure-aware. Traceability, IoT monitoring, cloning, extraction, quality control, assistant use, and evidence generation are all linked to controlled procedures. This makes the prototype suitable for future studies on digital quality assurance and GACP-informed workflows. We use *GACP-informed* rather than *GACP-compliant* deliberately: the platform was designed around GACP/GMP procedural requirements, but no formal regulatory audit or certification of the deployment has been carried out.

Deployment implications for small and medium producers. The reference deployment targets edge infrastructure. A Mini PC running Proxmox can host separate containers for the traceability platform application and MQTT broker, with optional containers for monitoring or auxiliary services. This deployment model supports local operation, reduced latency, independence from cloud infrastructure, and controlled maintenance. The approach is compatible with small and medium producers because it avoids mandatory cloud dependence and supports incremental adoption.

7.3. Connection with the clinical phase of PLANT2CLINIC

The agronomic module generates traceability and quality-context records intended to support a future hospital-pharmacy handoff. Their sufficiency for pharmacy reception has not been validated. The output of the prototype is not only a crop record, but a digitally documented batch passport, explicit S3 audit record, and handoff-ready dossier that accompanies the plant-derived material through extraction, quality control, transport, and hospital reception workflows.

As a roadmap element rather than a contribution of this article, the same architecture has been prototyped one stage downstream: a double-blind RCT clinical module (simulated data only) that receives the agronomic handoff dossier and reuses the S1–S2–S3 paradigm through three deterministic clinical agents (C-Trace, BIM, S2-C). Its purpose here is solely to show that the local-first, cryptographically anchored design extends to the clinical phase without architectural discontinuity; the module, its agents, and the standard blinding indices

it would report in a real trial [48, 49] are described in the Supplementary Material. This clinical prototype is not evaluated in the present work, and validation with real clinical data—informed by the operational context of the reference clinical site, *Hospital Público Andrés Ísola* (Puerto Madryn, Argentina)—is left entirely to future work.

7.4. Limitations

The historical S2 audit checks logged anomaly-band values and derives severity from recorded anomaly lists. It can miss omitted anomalies, and combined-condition levels are not independently recomputed. Its 100% agreement must therefore be interpreted as record-consistency evidence rather than a complete independent classifier validation. The archive covers approximately eleven weeks of agent logs, whereas the retained raw-reading table covers about three weeks; these windows are not interchangeable. The timing values come from a separate 100-case probe. Mobile demonstrations, replica operation, and healthcare interfaces have not undergone measured synchronization, interoperability, usability, or clinical evaluation.

This study has several limitations that should be considered when interpreting the reported results. First, the clinical phase of PLANT2CLINIC is outside the experimental scope of this article: no patient data, clinical outcomes, hospital dispensing events, or therapeutic results are analysed here. Second, the three clinical agents (C-Trace, BIM, S2-C) are presented as architectural models and prototype demonstrations on pseudonymised simulated data; they have not been evaluated on real patient cohorts, and no claims are made regarding their clinical diagnostic utility, sensitivity, specificity, or regulatory compliance. The BIM blinding integrity assessment in particular requires prospective validation in a real GCP-regulated trial context before any operational use. Third, the evaluation is preliminary and modest in size: the S1 question bank contains 100 queries, partitioned into 36 deterministic S1-1 lookups and 64 generative S1-2 queries, so each route is characterised on fewer cases than the bank total suggests. The corresponding Wilson 95% confidence intervals are therefore

wide—81.9%–98.5% for S1-1 raw matching (34/36), 90.4%–100% for S1-1 normalized (36/36), and 93.0%–99.4% for the pooled bank (98/100)—and the separate 90.2%–98.4% interval reported for the S2 robustness probe corresponds to its own 96/100 controlled responses, not to any S1 figure. A further caveat concerns the internal structure of the bank rather than its size: the 100 queries are not 100 independent questions. They derive from 62 distinct base questions, the remainder being paraphrase variants of a base question issued to probe robustness to surface wording. The deterministic route is unaffected—its 36 queries are 36 distinct questions—but the 64 generative S1-2 rows correspond to only 26 base questions, so its effective sample size is 26 rather than 64 and the corresponding Wilson interval widens from 94.3%–100% to 87.1%–100%. We report the paraphrase-aware interval as the conservative reading of the generative route. Larger and more diverse test sets—specifically, additional *base* questions rather than further paraphrases—are needed to establish statistical power for deployment decisions. Fourth, two of the reported quality figures rest on adjudication criteria whose strictness affects the number: the 4.7% deferral rate of the ablation is a marker count that yields a single genuine abstention under strict re-adjudication, and the panel's 100% factual fidelity becomes 96.9% (62/64) under a machine re-audit that penalises absence claims contradicted by the corpus. We report both readings rather than the more favourable one, and note that the stored responses are truncated at 400 characters, which limits any retrospective audit to the prefix in 55 of 64 cases. Relatedly, the S1 main-text correctness evaluation used a keyword exact-match heuristic, useful for checking factual grounding but insufficient for measuring explanation quality, usefulness, or safety; we address the retrieval side with retrieval-specific metrics (Hit@k, MRR, nDCG) for the retriever as an isolated component (Section 6.2, Supplementary Table S4), but generation-specific metrics remain to be reported separately in future studies. The deployment run also cannot support the abstention claim on its own: every query retrieved a non-empty evidence package, so the reported $\kappa = 1.00$ measures unanimity on a single observed category rather than the ability to discriminate deferrable from answerable cases. The

dedicated benchmark of Section 6.3 addresses that gap and bounds the claim: abstention is a property of the deterministic gate, not of the language model, and two limitations survive the gate fix—the gate keys on retrieval succeeding rather than on what was retrieved, so an off-topic but non-empty package still reaches the model, and contradictory evidence is not gated at all. That benchmark is itself small ($n = 12$ questions, single model) and partly marker-based; a larger set with human-adjudicated ground truth, and a gate predicate sensitive to evidence relevance rather than only to its presence, are the priority next steps. Fifth, the controlled retrieval ablation (Section 6.2) attributes the grounding and abstention behaviour of the S1-2 route to the retrieval component, but it also exposes a measurement limitation: the keyword exact-match proxy does not separate a grounded answer from a fabricated one (both arms scored 98.4%), so the safety-relevant contribution of retrieval is only visible through the auxiliary grounding and guardrail-activation metrics rather than through accuracy alone. A complementary comparison against a monolithic single-step RAG baseline (Section 6.2, Table 4) isolates the contribution of the S1-1 deterministic pre-filter and confirms that removing it degrades exact-value latency, canonical output form, and safety on factual look-ups; expert-scored faithfulness metrics that directly penalize fabrication nonetheless remain future work. Sixth, the S2 field-scale validation establishes deterministic fidelity and reproducibility against the published policy on real telemetry (54,982 checked room states from 55,266 records, with no discrepancies) but does not adjudicate the agronomic correctness of the threshold bands themselves, which would require expert-defined ground truth; the four input-validation exceptions surfaced by the complementary robustness probe identify an input-handling gap that must be resolved before production or regulated use. Seventh, a potential evaluator-bias risk exists because the system was developed, configured, and evaluated by the same team; this applies in particular to the three-member panel used for the qualitative S1-2 assessment, whose members were internal to the project rather than an independent external panel, so the high inter-rater agreement ($\kappa = 0.91$ and $\kappa = 1.00$) reflects internal consistency and should

not be read as independent confirmation. Blinded evaluation by external QA experts would strengthen the validity of the reported results. Eighth, S3 was validated as a workflow consistency layer rather than as a predictive or generative component, and its contribution to data integrity has not been measured quantitatively (e.g., hash verification latency, overhead per milestone). Ninth, the integrity layer is optional and should not be interpreted as proof of full regulatory compliance with GACP/GMP. Finally, the platform remains a research prototype and requires further validation, including field deployment over complete production cycles and formal regulatory assessment, before use in highly regulated production environments.

8. Conclusions and Future Work

PLANT2CLINIC implements an edge IoT architecture that connects environmental sensing, local processing, HMIs, QR-linked material identity, and human-reviewed audit dossiers. Medicinal cannabis serves as the reference crop for a plant-to-clinic information model in which production evidence can accompany material toward medicinal-use workflows. The operational contribution is the integration of these responsibilities into a locally deployed platform; generative AI extends access to evidence and explanation without determining alarm states or authorizing QA decisions.

An audit of 55,266 stored field evaluations found no discrepancies in 136,005 recorded anomaly-band decisions or 54,982 reconstructed room states, excluding 284 freshness states. A separate 100-case probe measured a median 0.24 ms per evaluation. This demonstrates policy fidelity within the evaluated data, not sensor accuracy, agronomic threshold validity, or network-wide real-time guarantees. The robustness probe produced 96 controlled responses and four input-validation exceptions, while seven audit workflows completed. The separate S1 experiments support fast deterministic evidence lookup and expose limitations of generative assistance, including sensitivity to misleading evidence and the need for an explicit abstention gate.

Future evaluation should prioritize complete cultivation cycles, sensor calibration and missing-data behaviour, gateway and network interruption, timestamp consistency, delivery semantics, recovery after outages, and concurrent workload at the edge. HMI studies should measure task completion, alarm interpretation, and traceability reconstruction with representative operators and QA reviewers. The cannabisIA and Health2Clinic demos support explanation of complementary workflows but do not substitute for these measurements. Integration should be tested with explicit batch-to-product-unit mappings, timestamped handoff events, and reconciliation after disconnects before claiming end-to-end clinical traceability. AI extensions require relevance-aware evidence gating and independent expert assessment. Pharmacy interoperability and clinical deployment require prospective validation: the downstream clinical prototype currently uses simulated data and supports no claim about therapeutic outcomes.

CRediT Authorship Contribution Statement

Francisco-Javier Ferrández-Pastor: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Manuel Platero-Horcajadas:** Software, Validation, Investigation, Data curation, Writing – review & editing. **Gregorio Bigatti:** Investigation, Resources, Writing – review & editing. **Mariana Lozada:** Investigation, Resources, Writing – review & editing. **Tomás Bosco:** Investigation, Resources, Data curation. **Irene McCarthy:** Investigation, Resources, Writing – review & editing. **Flavia Aragón:** Investigation, Resources. **Loreto Macía-Soler:** Conceptualization, Methodology, Supervision, Writing – review & editing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The PLANT2CLINIC Trazabilidad prototype described in this article was developed without commercial sponsorship related to the results reported here.

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Data Availability Statement

The Supplementary Material provides the aggregate and query-level results of the 100-question S1 evaluation and the 100-interaction S2 test log. The accompanying supplementary data archive includes the machine-readable S1 query records, summary and translation files, table-regeneration script, and the abstention and corrected-gate benchmark scripts and recorded outputs. Additional operational records are available from the corresponding author upon reasonable request. The SOP documentation used as the knowledge base is proprietary operational material; anonymised extracts are available from the corresponding author upon reasonable request. The platform prototype source code is available upon request subject to applicable licensing terms. A live demonstration of the platform is accessible at `https://trazabilidad.tecnologiashabilitadoras.cloud/` (access credentials can be requested through the corresponding author). This public instance is a cloud-hosted convenience deployment for reviewer and reader access; it does not contradict the local-first design, which refers to the intended production topology—the same application stack running on-premises at the edge (Mini PC, Proxmox) with local SQLite, MQTT broker, and Ollama inference, without mandatory cloud dependence. The benchmarks reported in this paper were obtained on the local edge configuration, not on the public cloud instance. The farmer video series is available at `https://agricultor.calidadgacp.com/videos/serie-argentina/`; the interactive farmer demonstrator is at `https://agricultor.calidadgacp.com/`. The healthcare demonstrator is at `https://pruebas.plant2clinic.com/`, with individual evaluation access available through the corresponding author. Credentials are not distributed in the manuscript. These mutable demo sites are not archival data repositories.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of the manuscript and submission materials, OpenAI Codex was used to assist with restructuring and editing the abstract, adapting the presentation to the target journal, preparing LaTeX and submission files, and drafting an explanatory architecture diagram. Interface screenshots were captured from the demonstrators without generative modification. The authors are responsible for reviewing and verifying the resulting text and take full responsibility for the content of the article. The local language models evaluated as research components are described separately in the Methods section.

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